DELTerra C

User Manual



V 1.6

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DOCUMENT INFORMATION

Document Information

Document Number	Document Title
V1.6	DELTerra C User Manual

Version Control

Rev.	Released Date	Change History	Author and Reviewer
V1.0	Oct. 16th, 2025		Written by Jessie Tseng, Viki Yeh. Reviewed by Jen Yang, Max Li, Vito Lin, Roy Lin, Herman Lin
V1.1	Nov. 22nd, 2025	**Disclaimer** • Revised disclaimer statement **1.3 Safety Considerations** • Newly added scetion for transport requirements • Added cabinet door closing illustration • Revised storage conditions **2.4 System Specification** • Revised noise **2.7 Fire Protection System** • Revised fire protection logic flowchart **3. Unpacking and Inspection** • Added key location • Added positioning accessories • Revised the quantity of PCS terminator **4. Installation** • Revised installation flowchart • Revised communication wiring instructions for two-set system • Revised main AC power cable qty. **4.1 Site Requirements** • Added illustration showing the distance between the cable outlet hole and the cable trench edge	Written by Jessie Tseng, Viki Yeh. Reviewed by Dora Hsiao, Hank Hsieh, Richard Lu, Avery Chen

		• Updated foundation design diagram for cabinets without an optional metal base **4.5 Metal Base Anchoring** • Updated illustration for positioning component connection **4.6 Cable Routing** • Revised cable routing steps for multiple cabinets in parallel **4.7 Cabinet Placement** • Added lifting cable length • Revised EMS IP address • Revised cabinet placement instructions • Updated alignment illustration for L-shaped brackets **4.9 Junction Box Installation** • Added bottom view of junction box **4.10 Cable Installation** • Added grounding notice and a step • Revised installation instructions for AC power cables, EMS communication cables, and auxiliary power cables **5.1 System Power-On** • Revised power-on sequence **5.2 Indicator Light Introduction** • Revised the descriptions for indicators **5.3 System Power-Off** • Revised power-off sequence **6.4 Maintenance Plan** • Revised inlet/outlet water pressure range of the liquid-cooling unit. **6.5 SOC Calibration** • Newly added section **6.6 Ventilation Louver Cleaning** • Newly added section. **6.7 Collant Replacement** • Newly added steps	
V1.2	Dec. 11th, 2025	**1.3 Safety Considerations** • Added battery storage requirements **2.4 System Specification** • Revised overload capacity **3 Unpacking Check** • Added a grouding cable to junction box kit • Added attached documents items to the unpacking check list **4.2 Preparation** • Revised the description of the main AC power cable quantity.	Written by Jessie Tseng, Reviewed by Max Li, Vito Lin

		• Added drill bit specification for expansion bolt installation. **4.9 Junction Box Installation** • Updated the dimension diagrams **4.10 Cable Installation** • Revised the installation instructions for the main AC power cables • Revised descriptions for aux. power cable installation method 1 & 2 • Revised the installation illustration for the auxiliary power cable (Method 1) **6.4 Maintenance Plan** • Added a mold inspection • Revised inlet/outlet water pressure range of the liquid-cooling unit. **6.6 Ventilation Louver Cleaning** • Added tool specifications • Updated figures **6.7 Coolant Replacement** • Added specifications for coolant replacment tools • Added a notice and illustrations for coolant filling step **Appendix 1 Grouding Architecture** • Newly added section	
V1.3	Jan. 6th, 2026	*The chapter sequence has been reordered.* **2.6 Power Conversion System** • Added description indicating that the PCS supports DRM0 **3.2 Preparation** • Added M10 and M12 terminal size specifications **4.1 Unpacking** • Added unpacking precautions **4.2 Unpacking Check** • Added grounding cable for the junction box • Revised AC power cable quantity for standalone cabinet with metal base to none **5.8 Cable Installation** • Updated the grounding cable instructions **7.5 SOC Calibration** • Added supplementary description for SOC calibration **7.8 Coolant Replacement** • Added coolant sepcification	Wirtten by Jessie Tseng

V1.4	Jan. 29th, 2026	**2.4 System Specification** • Added derating temperature for operation **3.2 Preparation List** • Removed external auxiliary power supply method **5.4 Cable Routing** • Adjusted cable try layout **5.8 Cable Installation** • Added grounding architecture diagrams • Removed external auxiliary power supply method • Added "– GND" suffix at the DI pin labels **Appendix 1 Grouding Architecture** • Removed and merged into Section 5.8	Wirtten by Jessie Tseng, Reviewed by Kevin Lin, Vito Lin, Max Li
V1.5	Mar, 3rd, 2026	**4.1 Unpacking** • Revised the junction box and metal base as optional items **5.4 Cable Routing** • Added cable routing instructions for configurations with external auxiliary power supply **5.8 Cable Installation** • Added grounding instructions between the cabinet and the metal base **6.1 Power-on Operation** • Revised Figure 6-1 to correct the MCCB OFF position. **7.8 Coolant Replacement** • Added coolant filling volume requirements	Wirtten by Jessie Tseng, Reviewed by Kevin Lin, Vito Lin, Max Li
V1.6	Mar. 5th, 2026	**2 Product Introduction** • Added notices for optional metal base and junction box accessories.	Wirtten by Jessie Tseng

ABOUT THIS MANUAL

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Synonym

L1/L2/L3/N	R/S/T/N

Abbreviation

MSD	Manual Service Disconnect
PCS	Power Conversion System
EMS	Energy Management System
BMS	Battery Management System
SPD	Surge Protective Device
SOC	State of Charge
AC	Alternating Current
DC	Direct Current
Comm.	Communication
Aux.	Auxiliary
Spec.	Specification

Disclaimer

Delta Electronics shall not be responsible for product warranty or legal liability for any property damage, personal injury, or other incidents resulting from the following circumstances:

1. Failure to properly install, operate, or maintain the product in accordance with this manual.
2. Loss or damage caused by improper battery operations, including over-discharge due to untimely system power-on or charging, frequent over-discharge resulting from improper maintenance, incorrect capacity expansion, or failure to fully charge the batteries for an extended period of time.
3. Installation, maintenance, or repair performed by non-authorized service providers or personnel.
4. Use of the product in environments that do not comply with the specifications in the manual (e.g., excessive temperature, humidity, strong magnetic fields, electrostatic interference, or corrosive gases), or exposure to conditions beyond the design specifications such as salt mist or corrosion (e.g., installation within 1 km of a coastal area).
5. Damage caused during transportation, handling (hoisting), storage, or unpacking due to improper practices.
6. Damage caused by improper cleaning methods (e.g., using chemical cleaners) is not covered under warranty scope.
7. Damage or malfunction caused by the use of unauthorized or non-original software, interface devices, parts, or consumables.
8. Damage or malfunction caused by natural disasters or other force majeure events (e.g., earthquakes, floods, fires).
9. Failure to install the product at a height sufficient to provide proper cable routing space and to address potential environmental risks such as flooding or snow accumulation, resulting in product damage or malfunction caused by water ingress or other environmental factors.
10. Relocation and reinstallation of the product to a non-original location without prior consent.
11. Opening the cabinet door under unsuitable environmental conditions.
12. Product serial number that has been altered, falsified, or failure to provide required information.
13. Normal wear and tear or aging, or surface defects, dents, or marks that do not affect product performance.
14. Minor noise or vibration that does not affect product performance.
15. The product or any of its components being stolen.
16. Environmental or safety incidents arising from failure to comply with relevant laws or regulations regarding use, recycle or disposal of product.
17. This product shall not be used in life-support systems, medical equipment, aerospace systems, or similar applications.
18. Delta Electronics shall not be liable for any consequential, indirect, incidental, or punitive damages arising from or in connection with this statement. In any event, Delta Electronics' maximum liability shall be limited to the actual amount paid by the customer for the purchased product.
19. Any disputes arising from this manual shall be governed by the laws of the Republic of China (Taiwan), with the Taipei District Court as the court of first instance.
20. The "Danger," "Warning," and "Caution" notes in this manual do not represent all required safety measures and are provided only as a supplement to other applicable safety precautions. Delta Electronics shall not be held responsible for any consequences resulting from failure to follow these or other applicable safety instructions.
21. Any damage to consumables (e.g., fuses, surge protectors).

The contents of this manual are subject to change without prior notice; please confirm with the your designated Delta Electronics representative that you are using the most up-to-date version obtained through sales channels before performing installation, operation, or maintenance.

1 IMPORTANT SAFETY INFORMATION

1.1 General Safety Statement

All installation, operation and maintenance procedures must comply with the applicable laws, regulations, codes, and standards of the location where the product is installed.

Before installation, operation and maintenance, please read through the entire manual and follow the procedures described in the manual. Unauthorized modification or alteration of the equipment, addition of components, or changes to the installation sequence are strictly prohibited. Neglecting these instructions may result in damage to the equipment, health hazards, or even loss of life. Please contact your designated Delta Electronics representative if any of the instructions are unclear.

1.2 Safety Warnings and Notices

This manual uses the following signs to highlight important information:



Indicates a hazardous situation which, if not avoided, **will** result in a death or serious injury.



Indicates a hazardous situation which, if not avoided, **could** result in death or serious injury.



Indicates a hazardous situation that, if not avoided, could damage the equipment and/or injury risk.



Indicates an important step or a helpful tip relevant to the current process or procedure.

DELTerra C includes the following labels that should be noted:



Properly grounding electrically-powered equipment to avoid the risk of fire and electric shock.



There is a risk of arc flash and electrical shock. It is prohibited to open the door during operation, and appropriate PPE and tools are required while working on this energized equipment.



Risk of danger. Proceed with care.



Flammable and explosive materials. Take fire and explosion prevention measures during operation.



Read the user manual to obtain equipment information.



Wear protective goggles.



Wear a safety helmet.



Wear protective gloves.



DO NOT step on the equipment to avoid damage.



DO NOT discard batteries at will. To avoid environmental pollution, batteries must be disposed of by authorized professionals.



Batteries are recyclable.

1.3 Safety Considerations

1.3.1 Transport Requirement

⚠CAUTION

- During transportation, the cabinet shall be properly secured to prevent tipping, sliding, or collision that may cause equipment damage.
- Handle the cabinet with care during loading and unloading to avoid impact or damage. During transport, protect it from severe vibration, direct sunlight, and rain exposure.
- For land transportation, the vehicle speed shall not exceed 80 km/h and must comply with local traffic regulations.
- Sudden braking and sharp turns are strictly prohibited during land transportation.

NOTICE

- During transportation of this product, compliance with applicable local regulations shall be confirmed, including whether relevant permits or approvals are required, and all national and local transportation regulations shall be followed.

1.3.2 Environment Requirement

⚠WARNING

- When the equipment is running, **DO NOT** block the vents and the cooling system. This will prevent damage to the equipment due to high temperature.
- **DO NOT** place the equipment in the following environments:
  - Near water sources or heat sources
  - Areas with dust, smoke, volatile gas
  - Locations with flammable/explosive substances, corrosive organic solvents
  - Areas with strong electromagnetic field interference
  - Areas with high salt

⚠CAUTION

- **DO NOT** install the system within 1 km of the coastline.
- When planning the site layout, noise levels shall be taken into account. The distance between the system and any building shall be no less than 3 meters. If required by local conditions, soundproof walls or equivalent noise-reduction measures shall be installed in compliance with all applicable local laws, regulations, and standards.

General Requirements

- The equipment must be installed in a clean, dry, and well-ventilated outdoor location.
- The equipment should be installed on a solid ground, not in low-lying areas (the water level of the site should be higher than the historical highest water level of this region).
- When the ambient temperature fails to meet the requirement, an external auxiliary power supply should be connected.

1.3.3 Storage Requirement

Table 1-1: Storage Requirement

≤1 year	0℃~35℃, <75% RH, no-condensing
≤1 month	-20℃~45℃,<75% RH, no-condensing

⚠CAUTION

It is recommended to place the battery into service in a timely manner. If the battery needs to be stored for an extended period, its SOC shall be maintained at 30%–50% during storage. The following procedures shall be performed according to the specified cycles; otherwise, the battery may be damaged:

① Every three months, perform a recharge cycle: first discharge the battery to 0% SOC, then recharge it to 30%–50% SOC for storage.

② Every six months, perform a charge–discharge cycle: first discharge the battery to 0% SOC, then fully charge it to 100% SOC, discharge it again to 0%, and finally recharge it to 30%–50% SOC for storage.

1.3.4 Personnel Safety

⚠ DANGER

- **DO NOT** install or remove cables while the power is on. Transient contact between the cable core and the conductor can cause electric arcs or sparks, which may lead to fire, explosion, or personal injury.
- In the event of system abnormalities such as sparks, smoke, or overheating, **DO NOT** approach the battery system.
- The voltage in some areas of the battery system is dangerous. **DO NOT** touch the live parts to prevent electric shock.

⚠WARNING

- During operations, use dedicated insulated tools to prevent electric shock or short circuits. The insulation withstand voltage level shall comply with the applicable local laws, standards, and regulatory requirements.
- **DO NOT** wear conductive items (e.g., watches, bracelets, rings, necklaces) during operation to prevent electric shock or burns.
- **DO NOT** power on the system before it is installed or confirmed by professionals.

General Requirements

- **DO NOT** operate the system alone.
- Appropriate personal protective equipment (PPE)—such as safety helmets, face shields, protective clothing, insulated gloves, and safety shoes—must be worn during operations, in accordance with local regulations.
- Before using any tools, ensure you understand the correct operating procedures to avoid injury or equipment damage.
- System installation, operation, and maintenance must be performed by trained and qualified personnel. These individuals must be fully aware of all safety precautions, familiar with applicable national/regional standards, and have undergone rigorous professional training. For special scenarios—such as electrical operations, work at heights, or handling specialized equipment—personnel must obtain approval from the relevant regulatory authorities in the country/region and strictly comply with local regulatory requirements.
- Only authorized personnel are allowed near the system; all other individuals must keep a safe distance.

1.3.5 Electrical Safety

⚠ DANGER

- Before connecting cables, make sure the equipment is intact and the cable connectors are properly installed. otherwise electric shock or fire may occur.
- Before performing any installation or maintenance, ensure that all power sources are completely shut off.
- **DO NOT** operate the system before the grounding conductor has been installed.

⚠ CAUTION

- Prevent foreign object from entering the equipment during operations, as this may result in equipment damage, load power derating, power failure, or personal injury.
- If any liquid is detected inside the equipment, disconnect the power supply immediately and **DO NOT** operate the system.

General Requirements

- All the electrical connection must meet the local electrical standards.
- Approval from the local power authority must be obtained before grid connection.
- When installing cables, tighten the screws in accordance with the torque requirements specified in this manual, and mark a torque mark (a straight line) at the junction of the locked components. If the screws become loose or have been disassembled, the torque marks are no longer intact, in this case, retighten the screws, reconfirm the torque and mark.

1.3.6 Mechanical Safety

⚠CAUTION

When closing the cabinet door, apply inward force at the lower part of the door to avoid damaging the lock. Ensure that the door's linkage is engaged with the black latch on the cabinet before performing the locking operation.

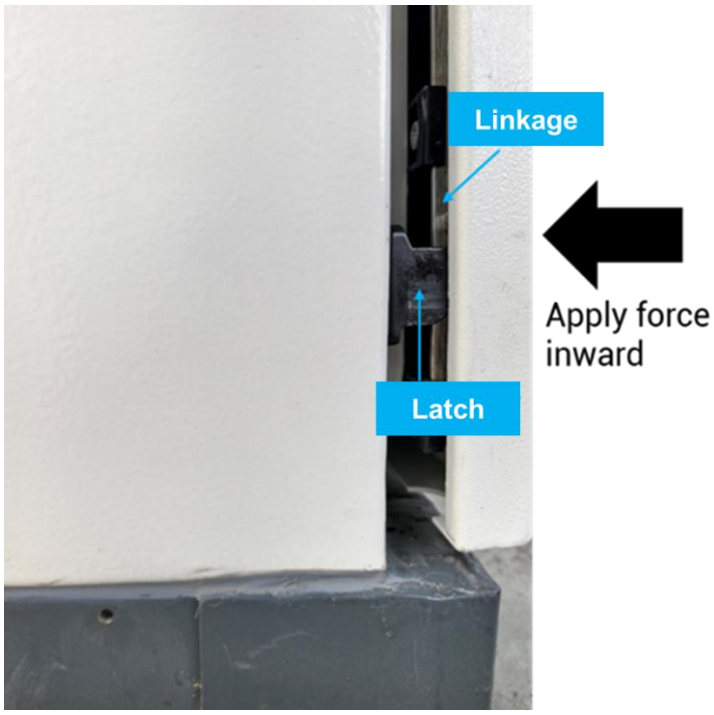


Figure 1-1: Cabinet Door Closing Illustration

General Requirements

- Repaint any scratches caused during transportation or installation in a timely manner. Equipment with scratches must not be exposed for an extended period.
- **DO NOT** install any additional components on the top of the system.

1.3.7 Battery Safety

⚠ DANGER

- **DO NOT** short-circuit the positive and negative terminals of the battery.

⚠WARNING

- Prevent the battery from mechanical vibration, falling down, collision, puncture of hard objects and pressure impact to avoid damage or fire.
- **DO NOT** use a damaged battery (such as damage caused when a battery is dropped, bumped, bulged, or dented on the enclosure), because the damage may cause electrolyte leakage or flammable gas release. In the case of electrolyte leakage or structural deformation, contact your designated Delta Electronics representative immediately to remove or replace the battery.
- When the battery malfunctions, its surface temperature may become extremely high. Avoid touching it to prevent burns.
- If batteries are exposed to water accidentally, **DO NOT** use them.
- **DO NOT** disassemble, modify or damage batteries.
- When it comes to electrolyte leakage, non-professional personnel are not allowed to approach it. The professionals must wear safety goggles, rubber gloves and protective clothing to prevent contact with liquid. If your eyes or skin accidently come into contact with electrolyte liquid, wash then with water immediately and ask for medical help.

⚠CAUTION

- Please use the standard model batteries provided by Delta Electronics. **DO NOT** mix them with other batteries, as this case may cause damage to batteries.
- The batteries must be operated within the specified charging rate or power limits. The charging voltage must not exceed the maximum specified in the technical requirements to prevent overcharging, which could impact battery performance and safety.

Recycling

- Dispose of waste batteries in accordance with local laws and regulations. **DO NOT** dispose of batteries as household waste. Improper disposal of batteries may result in environmental pollution or an explosion.
- If batteries are out of service life, contact a battery recycling company for disposal.
- **DO NOT** expose waste batteries to high temperatures or direct sunlight, or in environments with high humidity or corrosive substances

1.3.8 Precautions for Opening Door

The following table shows the conditions under which the cabinet door should not be opened. If it is necessary to open the door when the temperature and humidity conditions are not met, please contact the representative of Delta Electronics.

Table 1-2: Conditions Under Which the Door Should Not Be Opened

			
It is raining or snowing	Fog is noticeable or air contains a significant amount of sand particles	On-site humidity is above 95% (using a humidity sensor with temperature compensation and accuracy within 1%)	The minimum temperature of battery is below the dew point



Before opening the cabinet door, it is necessary to clean the accumulated water, ice, snow or other debris on the top to prevent the debris from falling into the equipment when opening the door.

2 PRODUCT INTRODUCTION

DELTerra C is an outdoor all-in-one Energy Storage System (ESS) designed for behind-the-meter commercial and industrial (C&I) applications. It integrates LFP battery packs, a Power Conversion System (PCS), a controller, a liquid cooling unit, and fire protection devices. Optional accessories, including the metal base and junction box, are available to facilitate installation.

NOTICE

The metal base and junction box (including the AC power cables between the junction box and cabinets) described throughout the manual are optional accessories. The instructions in this manual refer to those specific accessory models. Please refer to the purchase order for details about ordered accessories. The buyer is responsible for verifying that the AC junction box complies with local regulations.

2.1 Dimensions and Weight

Table 2-1: Dimensions and Weight

	Net Weight (excluding packaging)	**Gross Weight** (including packaging)	**Dimensions (W*D*H)** (excluding packaging)	**Dimensions (W*D*H)** (including packaging)
Cabinet	2720 kg	3010 kg	1000×1420×2250 mm	1030 x 1510×2270 mm
Metal Base	196 kg	281kg	1020×1420×220 mm	1150 x1550 x390 mm
Junction Box	63 kg	71 kg	1100 x 850 x 325 mm	1180 x 930 x 405 mm

Note: The gross weight of the metal base includes the weight of accessories.

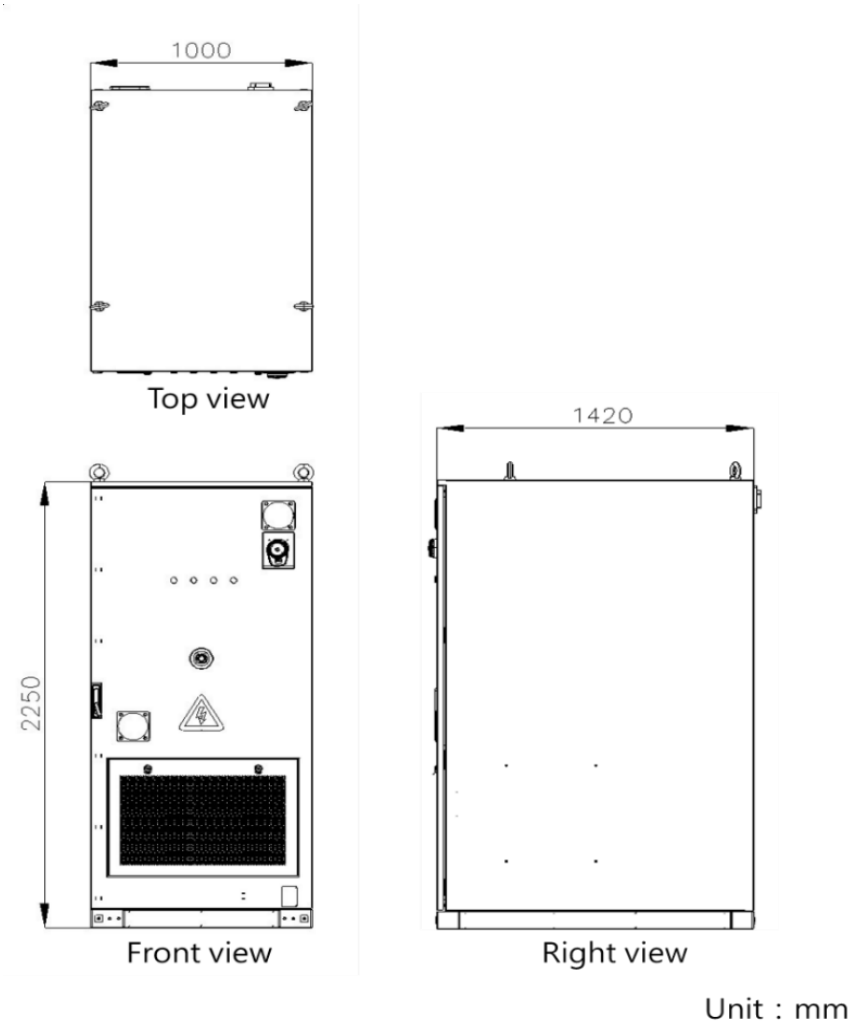
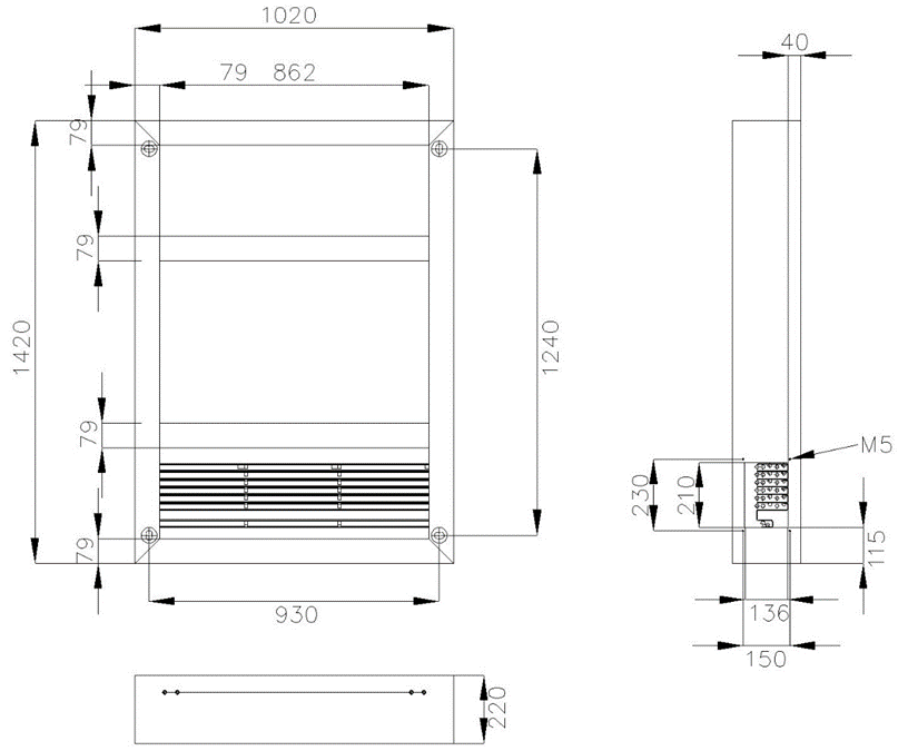
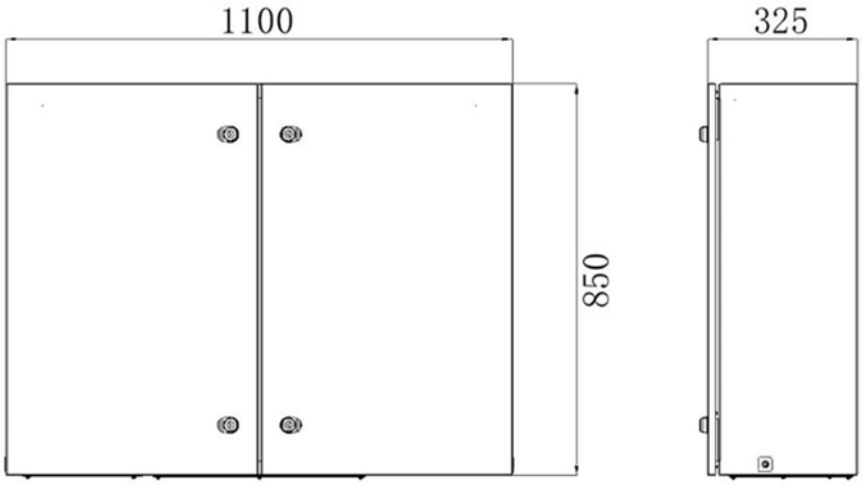


Figure 2-1: Cabinet Dimensions



Unit: mm

Figure 2-2: Metal Base Dimensions



Unit: mm

Figure 2-3: Junction Box Dimensions

2.2 Cabinet Layout

2.2.1 Exterior

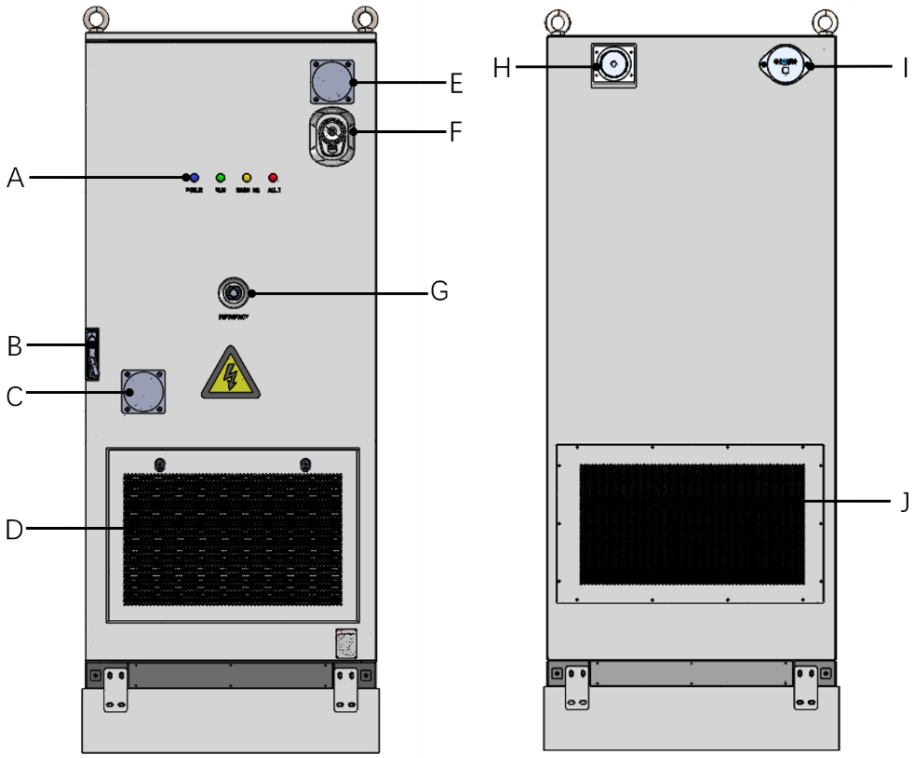


Figure 2-4: Cabinet Exterior Overview

Table 2-2: Cabinet Exterior Mapping Table

NO.	Item		NO.	Item
A	Indicators	– Power in blue – Run in green – Warning in yellow – Fault in red	F	Horn & strobe
B	Lock		G	EPO button
C	Gas-venting valve (inlet)		H	DN65 water inlet
D	Air inlet of liquid cooling unit		I	Explosion relief valve
E	Gas-venting valve (outlet)		J	Air outlet of liquid cooling unit

2.2.2 Interior

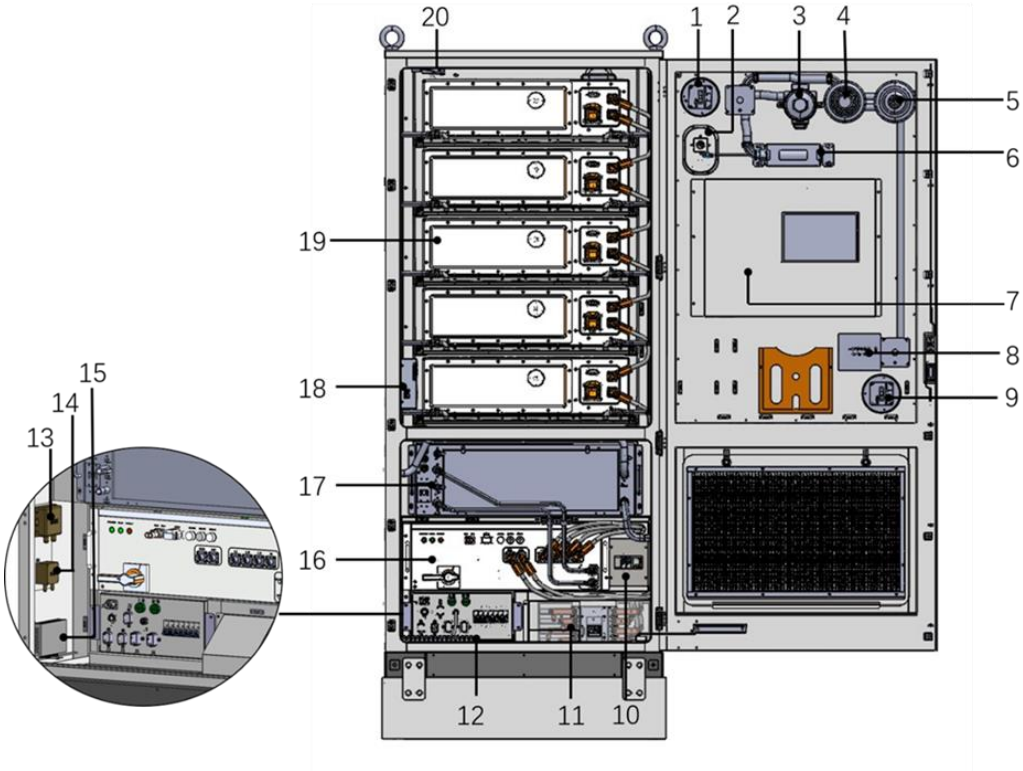


Figure 2-5: Cabinet Interior Overview

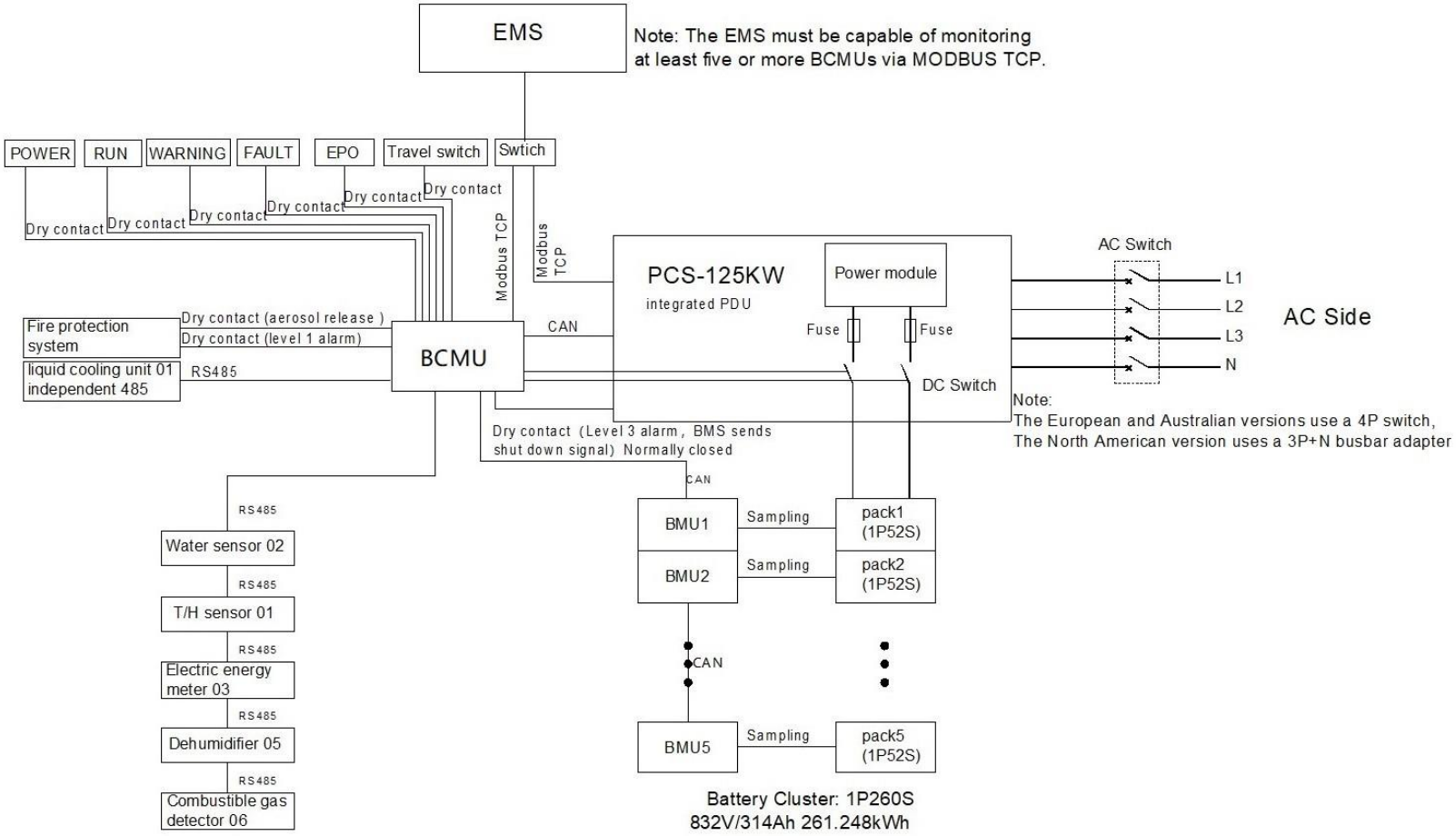
Table 2-3: Cabinet Interior Mapping Table

NO.	Item	NO.	Item
1	Gas-venting valve (outlet)	11	MCCB
2	Horn & strobe	12	Control box
3	Gas detector	13	T/H detector
4	Smoke detector	14	Flood sensor
5	Temperature detector	15	Switch
6	Aerosol	16	PCS
7	Controller	17	Liquid cooling unit
8	Fire relay box	18	Dehumidifier
9	Gas-venting valve (inlet)	19	Battery Pack
10	Power meter	20	Door sensor

NOTICE

For each set of ESS (1 ~ 5 cabinets connected in parallel), only the main cabinet is equipped with a controller. The controller integrates the BMS and PCS of each cabinet and acts as the central control unit for communication with EMS. For the controller installation, please refer to Appendix 1.

2.3 System Topology



Note: Fault levels are classified in ascending order of severity as Level 1, Level 2, and Level 3. Level 1 indicates a minor fault, while Level 3 represents a critical fault.

Figure 2-6: System Topology

2.4 System Specification

Table 2-4: System Technical Parameters

Specification		IEC / AS	UL
DC	Cell Type	LFP 3.2V / 314Ah	
	Battery Pack Configuration	1P52S / 52.2 kWh	
	Battery System Configuration	1P260S	
	Voltage Range	DC 728V~DC 936V	
	Installed Energy Capacity	261 kWh	
	Charge / Discharge Rate	≤0.5P	
AC	Rated Power	125 kW	
	Max. Power	137.5 kW	
	Rated Voltage	400V (-15%~+10%)	480V (-12%~+10%)
	Rated Current	180A	150A
	Max. Output Current	198A (Supports 110% rated power continuously)	165A (Supports 110% rated power continuously)
	Rated Frequency	50Hz / 60Hz	60Hz
	Overload Capacity	Supports 110% rated power continuously, and 120% rated power for 2 seconds.	
	Wiring Method	3P4W+PE	
	Thdi at Rated Power	< 3%	
	Power Factor / Adjustable Range	0.99 / -1~+1	

System	**Auxiliary Power**	≤6 kW
	Thermal Management	Liquid cooling / heating
	IP Rating	IP55
	Operating Temperature	-20℃~50℃, derating above 45℃
	Operating Humidity	0~95%RH, no-condensing
	Maximum Altitude	2000 m
	Noise	Front: 1 cabinet ≤ 66 dBA for 3m Rear: 1 cabinet ≤ 72 dBA for 3m
	Coolant	50% ethylene glycol–water solution, freezing point ≤ −35 °C

2.5 Battery Pack

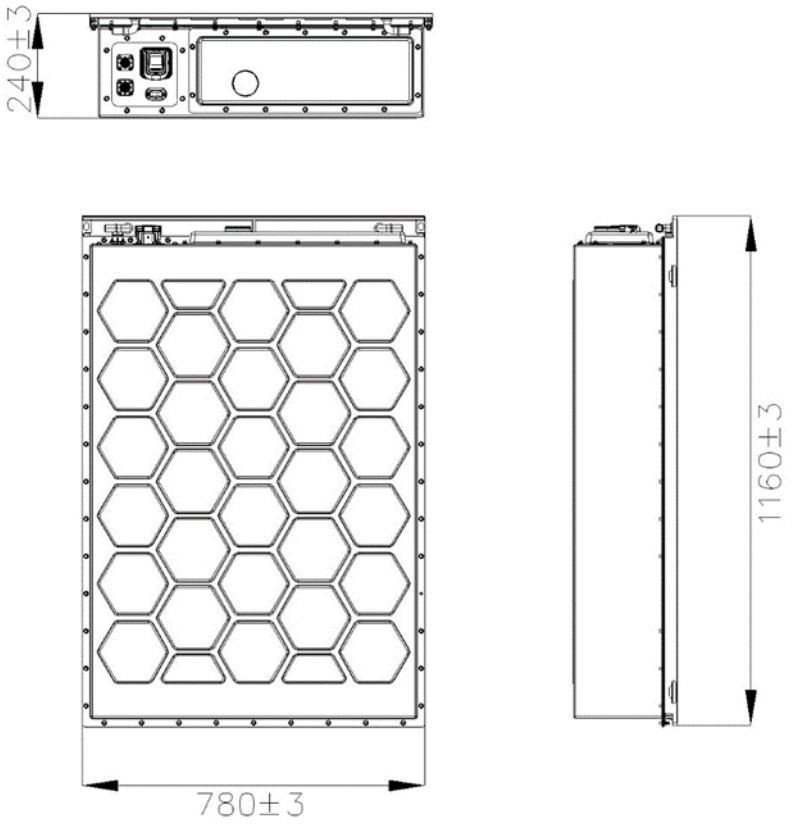


Figure 2-7: Battery Pack Dimensions

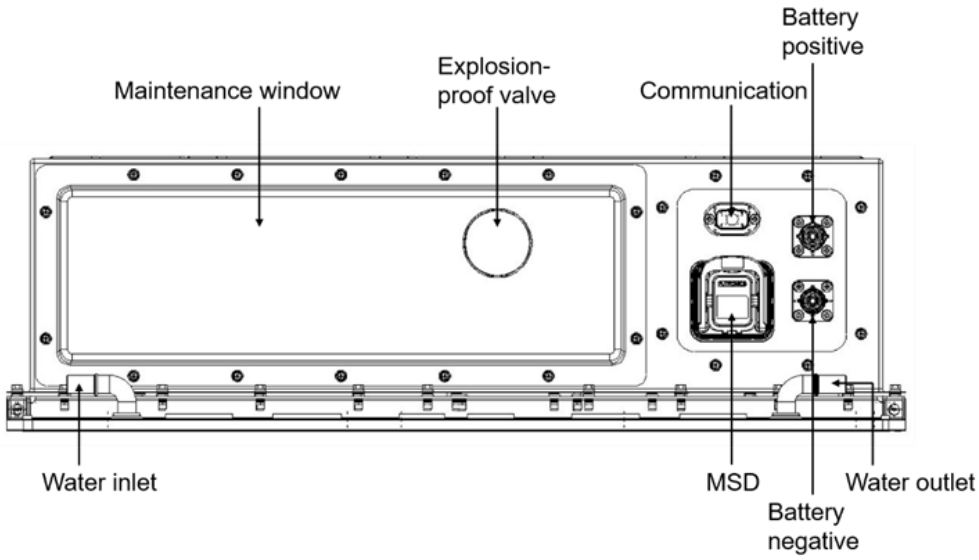


Figure 2-8: Battery Pack Interfaces

Table 2-5: Battery Pack Technical Parameters

1	**Configuration**	1P52S
2	**Nominal Energy**	52.2 kWh（0.5P@25±3℃）
3	**Nominal Voltage**	166.4V
4	**Voltage Range**	Module: 130V~189.8V Cell: 2.5V~3.65V
5	**Dimensions (W X D X H)**	780×1160×240 mm
6	**Weight**	345±5 kg
7	**IP Rating**	IP67
8	**Storage Requirement**	1 year: 0℃-35℃, < 70%RH 1 month: -20℃-45℃, < 70%RH

2.6 Power Conversion System

NOTICE

The PCS supports Demand Response Mode (DRM0) and can cease output upon receiving a stop command.

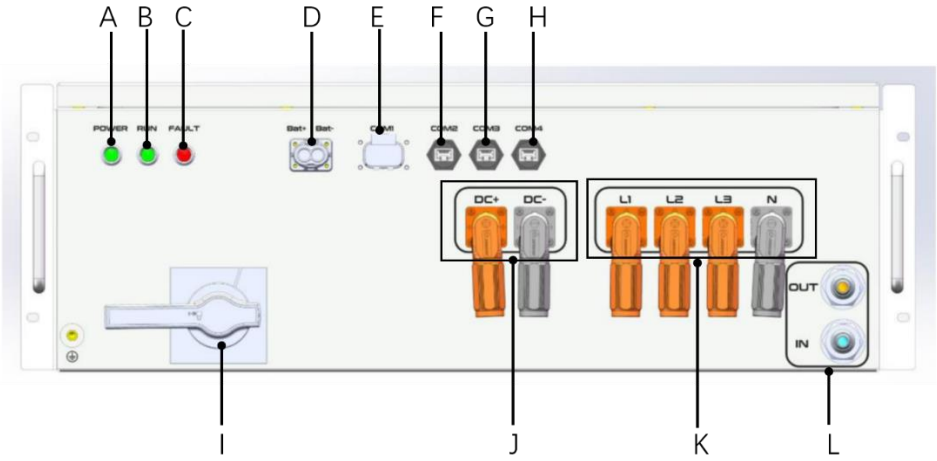


Figure 2-9: PCS Interface

Table 2-6: PCS Interface Mapping Table

NO.	Item	Description	NO.	Item	Description
A	POWER	DC side power-on indicator	**G**	COM3	Parallel communication port
B	RUN	Running indicator. When the system is in standby mode, the indicator blinks	**H**	COM4	EMS Ethernet port
C	FAULT	Fault indicator	**I**	Disconnector	Disconnecting battery high voltage
D	Bat+/Bat-	DC high voltage output ports	**J**	DC+/DC-	Connecting to battery rack B+/B-
E	COM1	Communication and DIO port	**K**	L1/L2/L3/N	AC side L1/L2/L3/N
F	COM2	Parallel communication port	**L**	OUT/IN	Outlet/inlet of liquid cooling unit

Table 2-7: PCS Technical Parameters

1	**Dimensions (W x D x H)**	680×850×240 mm
2	**Weight**	117.5 kg

2.7 Fire Protection System

This fire protection system consists of smoke, heat and gas detectors, a gas venting system, an explosion relief valve, an aerosol fire extinguisher, and a horn & strobe. It enables early fire detection and timely, effective suppression to minimize potential damage to the energy storage system.

Table 2-8: PCS Technical Parameters

1	**Working voltage**	DC 24V
2	**Fire extinguishing medium**	Aerosol
3	**Activation method**	Electrical Starting
4	**Agent release time**	15s (±5s)
5	**Operating environment**	-40℃~+54℃, ≤95%RH

Fire Protection Workflow

1 Level 1 Alarm: Triggered when either the heat or smoke detector reaches the Level 1 threshold. Then, the horn & strobe is activated, and the BMS receives the alarm signal.

2 Level 2 Alarm: Triggered when both the heat and smoke detectors reach the Level 2 threshold. Then, the fire extinguisher is activated, and the BMS receives the alarm signal.

3 Flammable Gas Alert: When flammable gas concentration reaches its threshold, the gas venting inlet and oulet valves open to release internal gas and allow fresh air intake.

4 When the above alarm threshold is reached, the PCS output power will be reduced to 0, and the system will stop charging and discharging.

5 When a fire alarm inside the cabinet is transmitted to the BMS, the BMS will output a fire alarm signal through the controller.

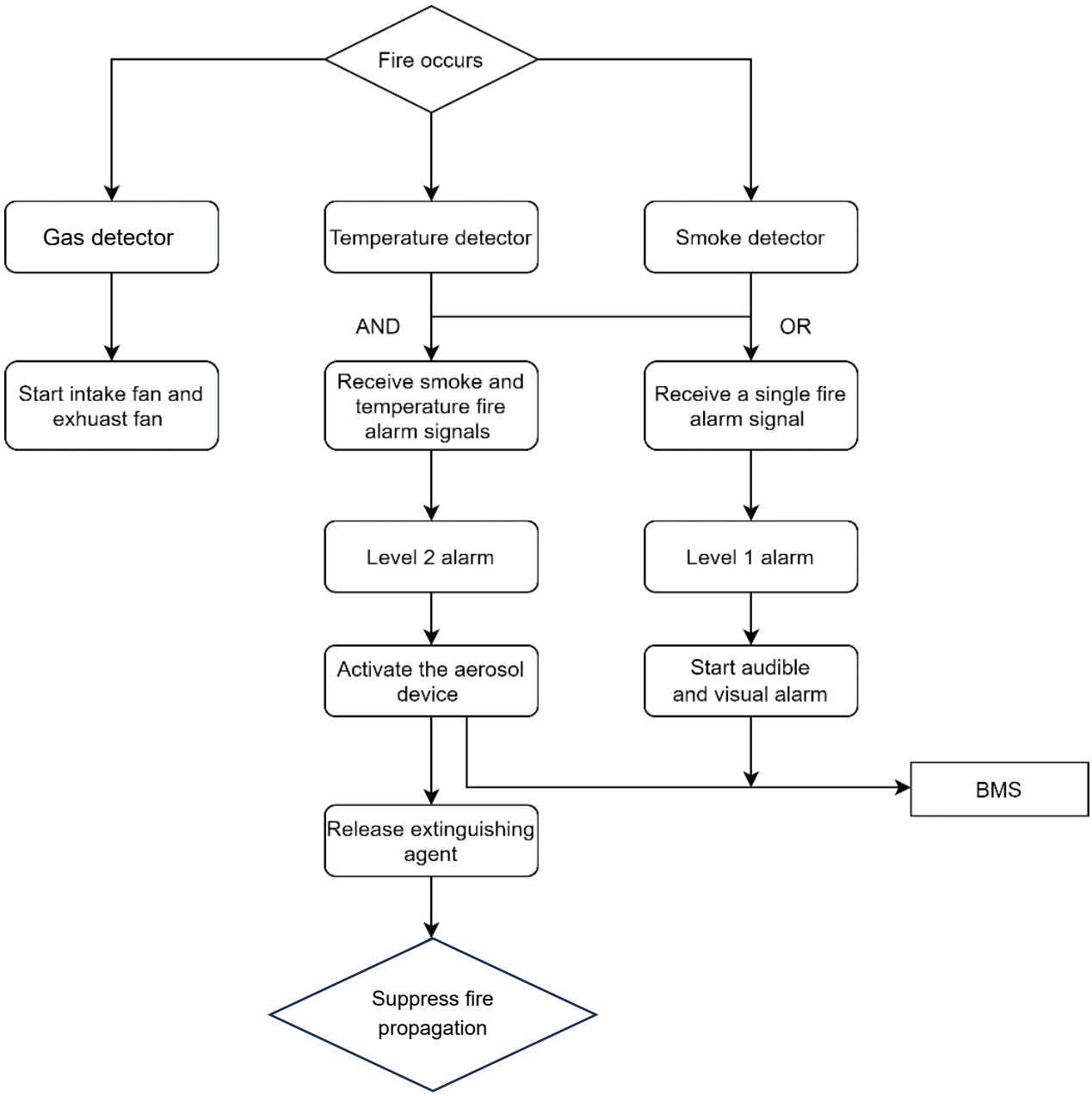


Figure 2-10: Fire Protection Logic Flowchart

3 PREPARATION

3.1 Recommended Tools

NOTICE

Due to varying site conditions, this tool list may not include all necessary tools. Installation personnel should prepare additional tools as needed based on the actual situation.

Table 3-1: Tool List

NO.	Tool	Specification
1	Crane / forklift	Load capacity > 3.5 T; Width of the forklift arm < 160 mm
2	Socket and torque wrench	Range: 0~180 N·m
3	Screwdriver	Phillips, Slotted
4	Wire stripper	Conventional
5	Crimping pliers	Conventional
6	Hot air gun	Conventional
7	Power drill and drill bit	 (Unit: mm) A power drill as specified above and a Φ18 mm drill bit
8	Hammer	Conventional
9	Marker	Conventional
10	Ink line marker	Conventional
11	Multimeter	Range up to 1000 VDC
12	Safety protection tools (PPE)	including, but not limited to, insulated gloves, insulated shoes, safety helmets, arc-flash protective clothing, and other personal protective equipment (PPE) as required.

3.2 Preparation List

⚠CAUTION

- An isolating switch or circuit breaker compliant with UL 489 or other applicable standards shall be installed on the AC power supply side.
- There is no SPD provided on AC side inside DELTerra C. It is recommended to install SPDs, or determine the necessity in accordance with local regulations and applicable standards. If auxiliary power cables are exposed to lightning or induced overvoltage risks, an SPD meeting the specifications in the table below shall be installed to ensure system safety and protection effectiveness.

Table 3-2: SPD Specifications

<table>
<tr><th></th><th>IEC/AS Version</th><th>UL Version</th></tr>
<tr><td>Model</td><td>DEHN 952 205</td><td>DENN 908 341</td></tr>
<tr><td>Input Voltage</td><td>1P2W, 230Vac</td><td>3P4W, 277/480Vac</td></tr>
<tr><td>Requirement</td><td colspan="2">The overvoltage category is reduced to OVC II.</td></tr>
<tr><td>Reference</td><td>
<table>
<tr><td>Type DG M ...</td><td>TN 275 FM</td></tr>
<tr><td>Part No.</td><td>952 205</td></tr>
<tr><td>SPD according to EN 61643-11 / IEC 61643-11</td><td>type 2 / class II</td></tr>
<tr><td>Max. continuous operating voltage (a.c.) (U_C)</td><td>275 V (50 / 60 Hz)</td></tr>
<tr><td>Nominal discharge current (8/20 µs) (I_n)</td><td>20 kA</td></tr>
<tr><td>Max. discharge current (8/20 µs) (I_{max})</td><td>40 kA</td></tr>
<tr><td>Voltage protection level [L-PE]/[N-PE] (U_P)</td><td>≤ 1.5 / ≤ 1.5 kV</td></tr>
<tr><td>Max. mains-side overcurrent protection</td><td>125 A gG</td></tr>
<tr><td>Approvals</td><td>KEMA, VDE, UL</td></tr>
<tr><td>Type of remote signalling contact</td><td>changeover contact</td></tr>
</table>
</td><td>
<table>
<tr><td>Type DG MU ...</td><td>3PY 480 4W+G</td></tr>
<tr><td>Part No.</td><td>908 341</td></tr>
<tr><td>SPD classification acc. to ANSI/UL 1449 4th Ed.</td><td>Open-Type 1 SPD</td></tr>
<tr><td>SPD classification acc. to CSA - C22.2 No. 269.1-14</td><td>Type 4-1 Component Assembly</td></tr>
<tr><td>Max. continuous operating voltage AC [L-N] / [L-G] / [L-L] / [N-G] (MCOV)</td><td>385 V a.c. / 565 V a.c. / 770 V a.c./ 180 V a.c.</td></tr>
<tr><td>Nominal discharge current (8x 20 µs) (I_n)</td><td>20 kA</td></tr>
<tr><td>Max. discharge current (8/20) (I_{max})</td><td>50 kA</td></tr>
<tr><td>Voltage Protection Rating [L-N] / [L-G] / [L-L] / [N-G] (VPR)</td><td>1200 V_{pk} / 1800 V_{pk} / 2500 V_{pk} / 600 V_{pk}</td></tr>
<tr><td>Max. mains-side overcurrent protection</td><td>Not needed</td></tr>
<tr><td>Approvals</td><td>UL, CSA</td></tr>
</table>
</td></tr>
</table>

NOTICE

- All user-prepared cables must comply with local laws, regulations, and applicable technical standards.
- Cable selection, routing, and connection must comply with local laws, regulations, and applicable technical standards.
- Before installing power cables, ensure the cable labels are correctly identified and that all terminals are properly insulated.
- In humid environments where snakes, rodents, or other small animals may be present and could damage the cables, it is recommended to install metal conduits to enhance the level of protection.
- The cabinet does not include a built-in UPS. In the event of a utility power outage, the auxiliary power supply is provided by the battery system; if an external auxiliary power supply is required, please contact Delta for further evaluation and discussion.
- For communication between two sets of parallel systems (e.g., 5+5 configuration), each master cabinet shall be connected to the WAN switch via one communication cable, and the WAN switch shall then be connected to the EMS server through a single communication cable.

Table 3-3: Cable List – With Metal Base

Item	Cable Type	Cable Spec.	QTY – Standalone Cabinet	QTY – Multiple Cabinets in Parallel (N=cabinet quantity; up to 5 cabinets in parallel as a set)	Scope	
1	AC Power Cable with cable lug suitable for M10 bolts *(cabinet → junction box)*	95mm^2 – XLPE – 600V (Withstand current greater than 230A at ambient temperature below 40°C)	4 (prepared by user)	N×4 (provided by Delta)	Standalone	User-prepared
					Multiple	Delta-provided
2	Comm. Cable for PCS interconnection	CAT6A-SFTP cable with shielded RJ45 connector	None	1 between two cabinets / Total quantity N-1	Delta-provided	
3	Comm. Cable for switch interconnection	CAT6A-SFTP cable with shielded RJ45 connector	None	1 between two cabinets / Total quantity N-1		
4	Comm. Cable to EMS	CAT6A-SFTP cable with shielded RJ45 connector	1	1 per set	User-prepared	
5	Main AC Power Cable with cable lug suitable for M12 bolts *(junction box → switchboard)*	240mm^2 – XLPE – 600V	None	A maximum of three sets (L1/L2/L3/N) can be configured. The cable specs and quantities shall be selected based on local electrical regulations, system requirements, and installation conditions. The dimensions of the junction box are provided in Section 4.9		

6	Grounding Cable	In accordance with the grounding architecture design of the site (refer to Section 5.8.1).		User-prepared
7	(Optional) Signal Cable	0.5mm^2 – 600V – 2-core twisted and shielded cable	Subject to the user's design	

Hole	B (Max.)	L	D	d1	d2
M10	26	60	18	13.5	10.5
M12	47	Select the specifications according to the cable used.			

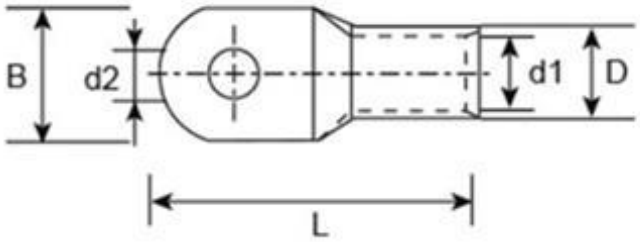


Figure 3-1: Specification Requirements for M10 & M12 Cable Lugs

Table 3-4: Cable List – Without Metal Base

Item	Cable Type	Cable Spec.	QTY – Standalone Cabinet	QTY – Multiple Cabinets in Parallel (N=cabinet quantity; up to 5 cabinets in parallel as a set)	Scope
1	AC Power Cable with cable lug suitable for M10 bolts	95mm^2 – XLPE – 600V (Withstand current greater than 230A at ambient temperature below 40°C)	N×4		User-prepared
2	Comm. Cable for PCS interconnection	CAT6A-SFTP cable with shielded RJ45 connector	None	1 between two cabinets / Total quantity N-1	
3	Comm. Cable for switch interconnection	CAT6A-SFTP cable with shielded RJ45 connector	None	1 between two cabinets / Total quantity N-1	
4	Comm. Cable to EMS	CAT6A-SFTP cable with shielded RJ45 connector	1	1 per set	
5	Grounding Cable	In accordance with the grounding architecture design of the site (refer to Section 5.8.1).			
6	(Optional) Signal Cable	0.5mm^2 – 600V – 2-core twisted and shielded cable	Subject to the user's design		User-prepared

4 UNPACKING

NOTICE

Before unpacking, please check whether the packaging has been subjected to impact or damaged. If any abnormality is found, please take photos for record and notify the supplier immediately.

NOTICE

The base drilling template plate is included in the carton packaging. Please pay special attention during unpacking and retain it to avoid accidental disposal.

4.1 Unpacking

Cabinet

The cabinet is protected with EPE foam and stretch film. When removing the packaging, please follow the steps below:

STEP 1: Cut the packing straps.

STEP 2: Tear or cut open the stretch film.

STEP 3: Remove the attached baffle and the surrounding EPE foam.



Figure 4-1: Cabinet Packaging

Metal Base

The metal base is protected with a wooden crate and pallet. A positioning drilling template is included inside the packaging. Handle with care during unpacking to avoid damaging the positioning cardboard.

STEP 1: Cut the outer packing straps of the wooden crate and remove the crate.

STEP 2: Cut the internal straps securing the base, then remove the drilling template for use during installation positioning and drilling.

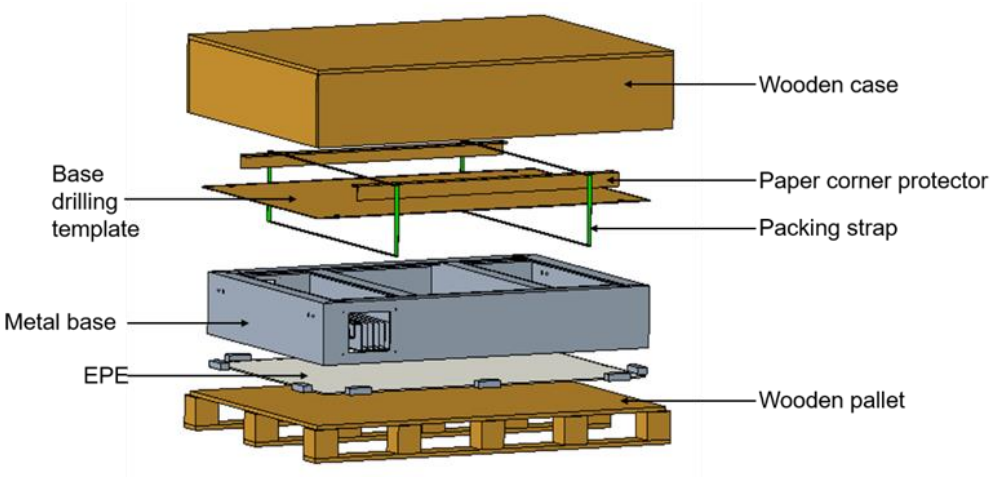


Figure 4-2: Exploded View of Metal Base Packaging

4.2 Unpacking Check

NOTICE

After unpacking, verify the product model and confirm that all equipment is complete in accordance with the accompanying packing list. In case of any discrepancy or damage, please take photos for record and notify the supplier immediately.

NOTICE

The key locations for each cabinet and junction box are as follows. Please keep them properly:

① Cabinet: Two (2) door keys and one (1) vent key, preset and hung on the door lock.

② Junction box: One (1) door key and one (1) grounding wire, packed together in a transparent bag.

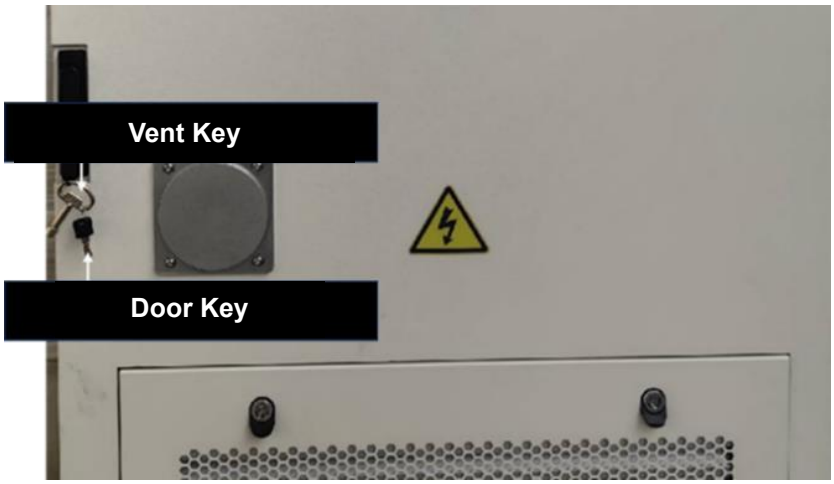


Figure 4-3: Key Location

Table 4-1: Package List – Cabinet

NO.	Item	Stand-alone Cabinet	**Multiple Cabinets in Parallel** (N=cabinet quantity; up to 5 cabinets in parallel as a set)	Remark
1	All-in-one cabinet	1 unit	1 unit × N	The main components inside the cabinet have been installed.
2	MSD	5 pcs	5 pcs × N	To be installed on the battery packs
3	Fork pocket cover	2 pcs	2 pcs × N	To be installed on both front and rear sides of the cabinet bottom
4	Fork pocket cover kit	1 set	1 set x N	One set includes 16 × M4 bolts
5	Waterproof plug	4 pcs	4 pcs × N	To be installed on the top of the cabinet after removing the lifting rings
6	PCS comm. terminator	1 pc	1 pc x N	To be installed on the comm. ports of the first and last cabinet
7	Aux. power cable	1 set	1 set × N	To be used only if external auxiliary power (Method 2) is required. Spec.: 2mm²-XLPE-600V-2m
8	Concrete foundation kit	1 set	1 set × N	To be used only if the metal base is not selected. One set includes: – 4 x L-shaped brackets – 4 x square beveled washers – 4 x M16 expansion bolts – 8 x M10 bolts
9	Document	2 pcs	2 pcs	– 1 x factory inspection report – 1 x shipping inspection checklist

Table 4-2: Package List - Junction Box (Optional)

NO.	Item	Standalone Cabinet	**Multiple Cabinets in Parallel** (N=cabinet quantity; up to 5 cabinets in parallel as a set)	Remark
1	Junction box kit	None	1 set (per set)	One set includes: – 1 × junction box – 8 × M6 bolts – 1 x grounding cable

Table 4-3: Package List - Metal Base (Optional)

NO.	Item	Standalone Cabinet	**Multiple Cabinets in Parallel** (N=cabinet quantity; up to 5 cabinets in parallel as a set)	Remark
1	Metal base kit	1 set	1 set × N	One set includes: – 1 × metal base – 4 × mounting bracket – 4 × M16*120 expansion bolts – 16 × M10 bolts
2	Positioning guide	2 pcs	Total quantity 1-2 cabinets: 2 pcs Total quantity ≥3 cabinets: N × 20% (integer only) × 2	Used as temporary positioning aids, fixed onto the base during cabinet installation
3	Cover plate kit	2 pcs	1 set (per set)	<u>Standalone Cabinet</u> one set includes: – 2 × cover plates – 8 × M5 bolts. <u>Multiple Cabinets</u> one set includes: – 2 × cover plates – 12 × M5 bolts.
4	AC power cable	None	4 pcs x N	95 mm^2 – XLPE – 600V
5	Comm. cable	None	(N-1)*2 pcs	CAT6A-SFTP cable with shielded RJ45 connector for PCS & switch interconnection

5 INSTALLATION GUIDE

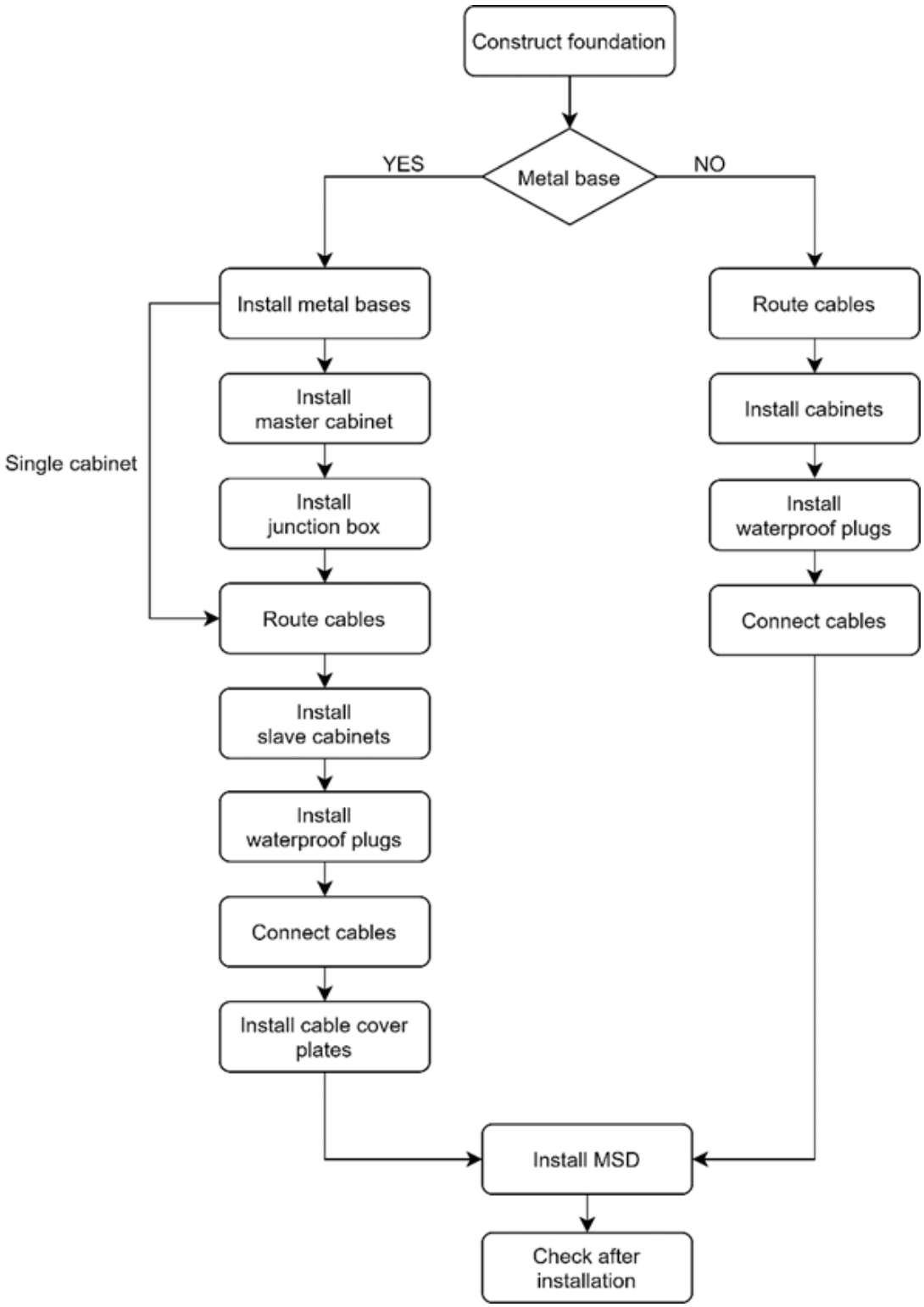


Figure 5-1: Installation Flowchart

5.1 Site Requirements

⚠CAUTION

- **DO NOT** install the system within 1 km of the coastline.
- Noise levels shall be considered when planning the site layout. The system shall be installed at least 3 meters away from any building. Where required by local conditions, soundproof walls or equivalent noise-reduction measures shall be provided in accordance with all applicable local laws, regulations, and standards.

5.1.1 Foundation Requirements

⚠CAUTION

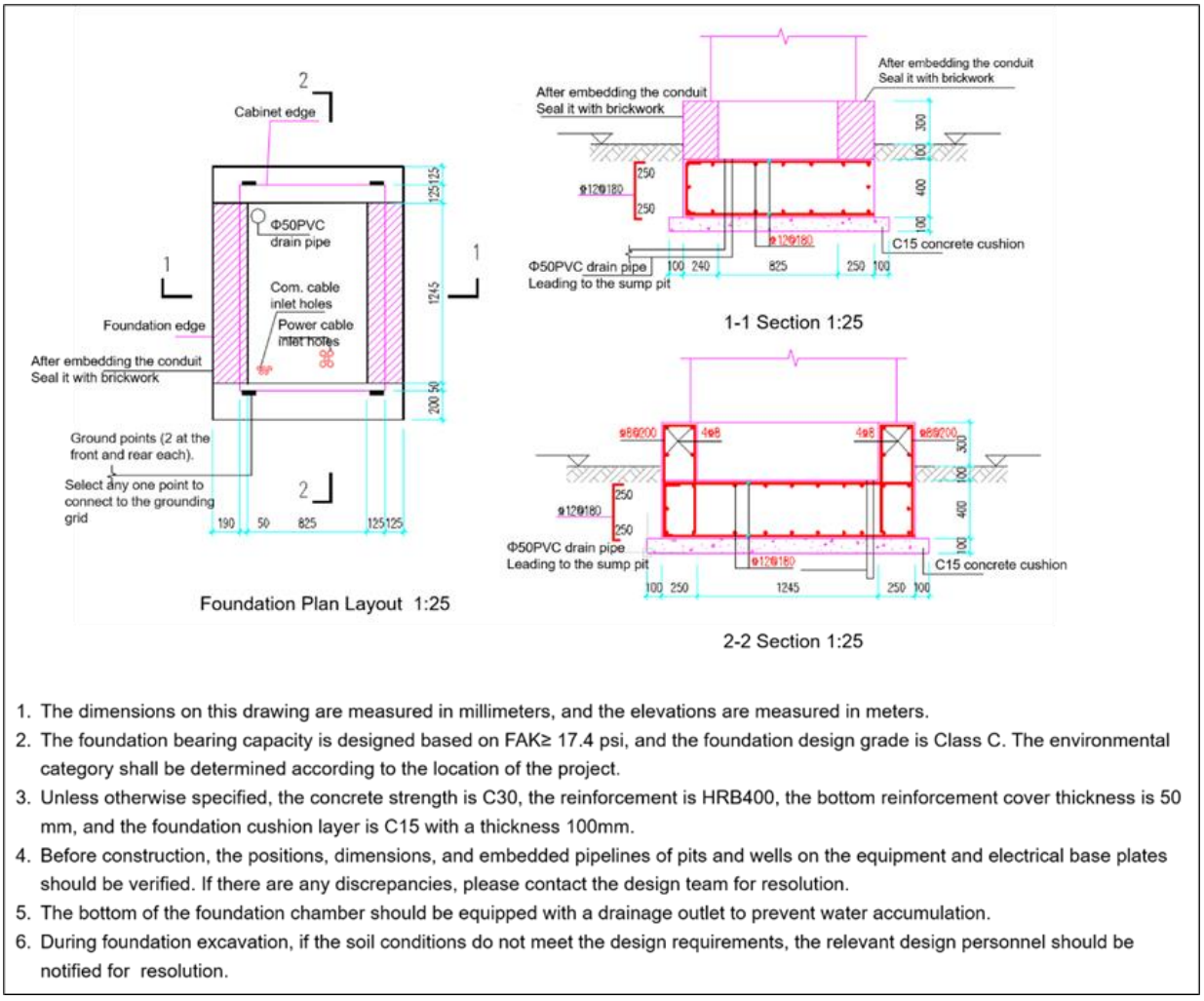
- Consult with local civil / structural and fire protection engineers to ensure that the foundation design complies with all relevant regulations. Key considerations include drainage slope, drainage ditches, and firewater collection and discharge systems to ensure structural integrity and proper waste water management.
- Foundation construction shall only proceed after the civil / structural and electrical design drawings have been fully reviewed and approved by the respective licensed engineers, and further validated by Delta Electronics.
- Before construction, ensure that the grounding grid, drainage infrastructure, and fire protection facilities are properly planned and designed.

5.1.1.1 Foundation Requirements with the Metal Base

- The base foundation must be properly compacted and leveled. The soil bearing capacity must meet **FAK ≥ 17.4 psi**.
- Concrete strength shall be **≥ 210 kgf/cm² (3,000 psi)**, with a minimum **thickness of 130 mm.**
- The allowable surface **flatness deviation** is **4 mm per 2 meters**.
- The foundation should extend at least **165 mm** beyond the perimeter of the metal base on all sides.

5.1.1.2 Foundation Requirements without the Metal Base

- The base foundation must be properly compacted and leveled. The soil bearing capacity must meet **FAK ≥ 17.4 psi.**
- Concrete strength shall be **≥ 210 kgf/cm² (3,000 psi)**, with a minimum **thickness of 400 mm.**
- The allowable surface **flatness deviation** is **4 mm per 2 meters**.
- The foundation must be raised at least **300 mm** above ground level and extend **125 mm** larger beyond the perimeter of the cabinet base to ensure the structural stability.
- It is necessary to consider the cable outlet and reserve the trench in advance.



1. The dimensions on this drawing are measured in millimeters, and the elevations are measured in meters.
2. The foundation bearing capacity is designed based on FAK≥ 17.4 psi, and the foundation design grade is Class C. The environmental category shall be determined according to the location of the project.
3. Unless otherwise specified, the concrete strength is C30, the reinforcement is HRB400, the bottom reinforcement cover thickness is 50 mm, and the foundation cushion layer is C15 with a thickness 100mm.
4. Before construction, the positions, dimensions, and embedded pipelines of pits and wells on the equipment and electrical base plates should be verified. If there are any discrepancies, please contact the design team for resolution.
5. The bottom of the foundation chamber should be equipped with a drainage outlet to prevent water accumulation.
6. During foundation excavation, if the soil conditions do not meet the design requirements, the relevant design personnel should be notified for resolution.

Figure 5-2: Foundation Design Drawing (Without Metal Base - Reference Only)

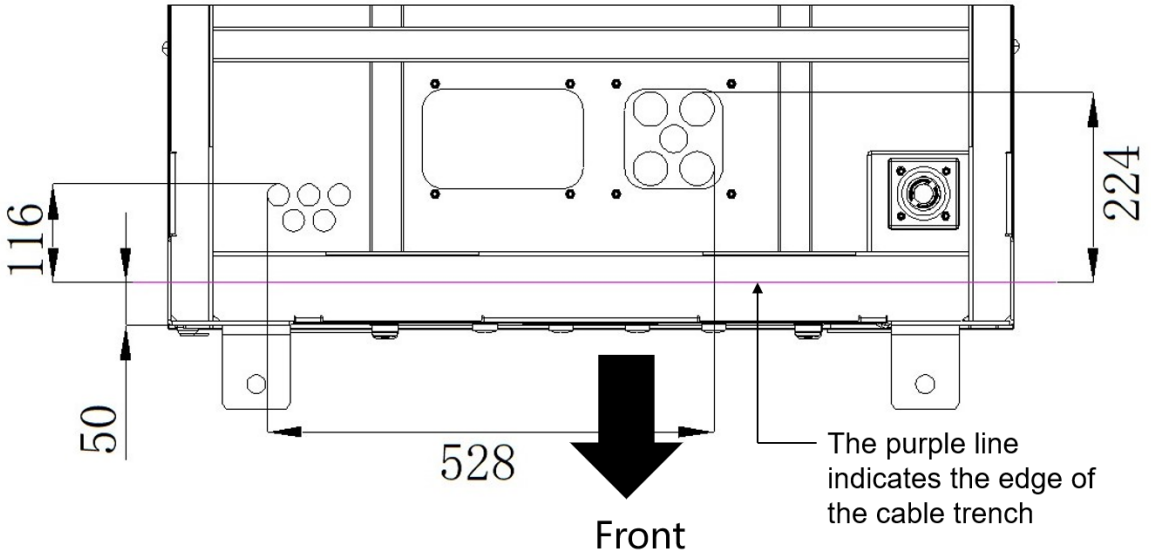


Figure 5-3: Cable Outlet to Cable Trench Edge Distance Illustration

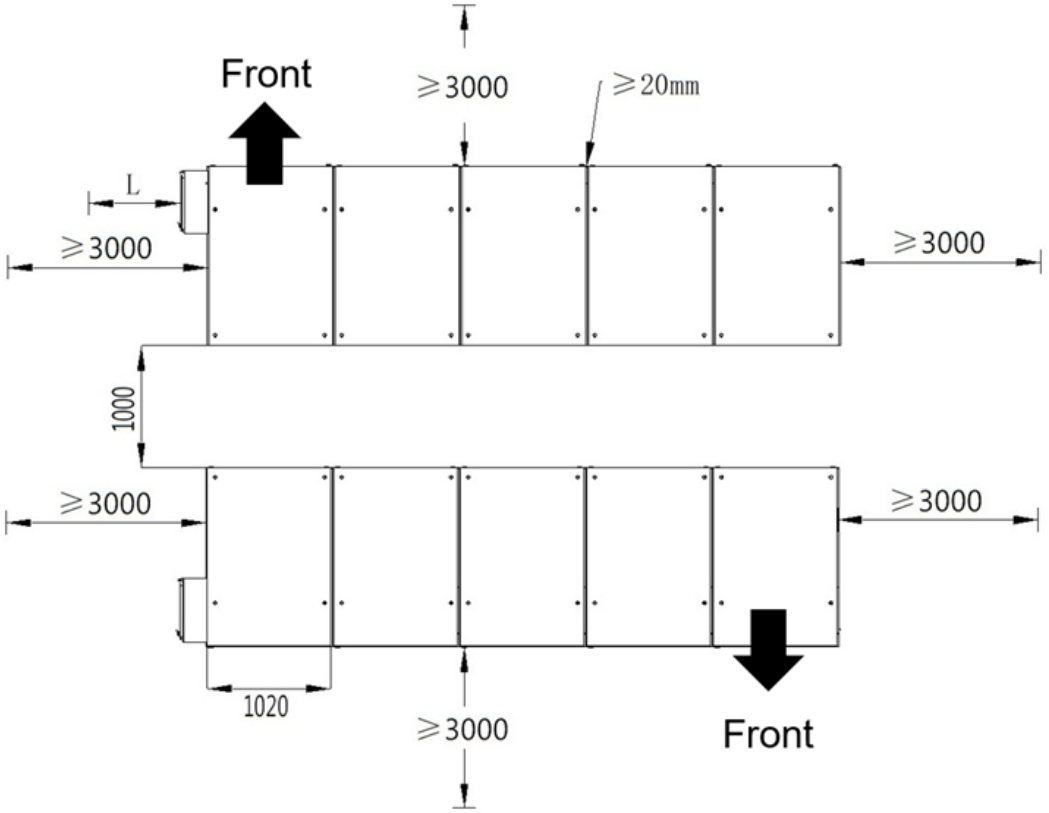
5.1.2 Clearance Requirements

Table 5-1: Clearance Requirements

Direction	Minimum Clearance	
	Single Cabinet	Multiple Cabinets
Between Cabinets	NA	*≥20mm
Front of Cabinet	≥3000mm	
Rear of Cabinet	1 parallel: ≥500mm	1 parallel: ≥500mm 2 parallels (back-to-back): ≥1000mm
Both Sides of Cabinet	Recommendation: ≥3000mm, Adjustments may be made in accordance with local fire codes.	
L	≥1000mm	

NOTICE

*Since the metal base is 20 mm wider than the bottom of the cabinet, no additional 20 mm spacing is required between cabinets when using the metal base configuration.



Unit: mm

Figure 5-4: Clearance Requirements

5.2 Positioning

5.2.1 With Metal Base – Bolt Positioning and Pre-Embedding

STEP 1: Use an ink line marker to draw a reference line ① on the horizontal axis of the foundation. Then, draw a reference line ② on the vertical axis on one side of the foundation, ensuring that the two lines are perpendicular to each other.

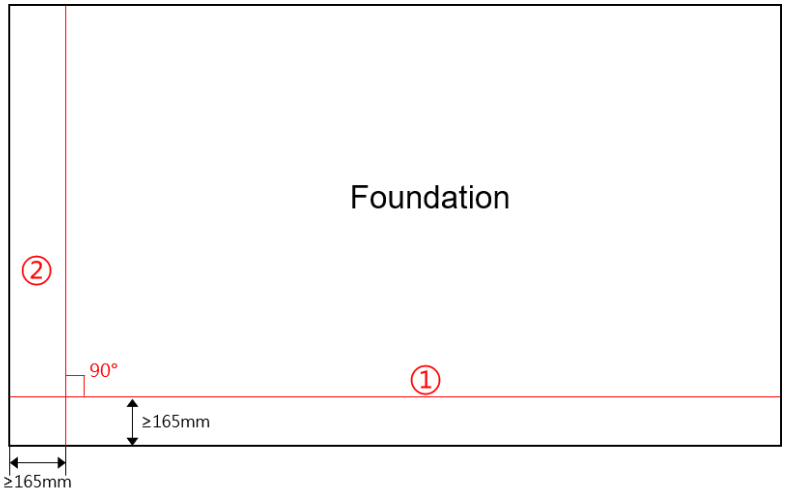


Figure 5-5: Metal base Positioning – Drawing Lines

STEP 2: Align one corner of the drilling template with the intersection of the two reference lines (as indicated by the blue marker in the figure below). Mark the positions of the four drilling points (labeled as positions A, B, C, and D in the figure below)

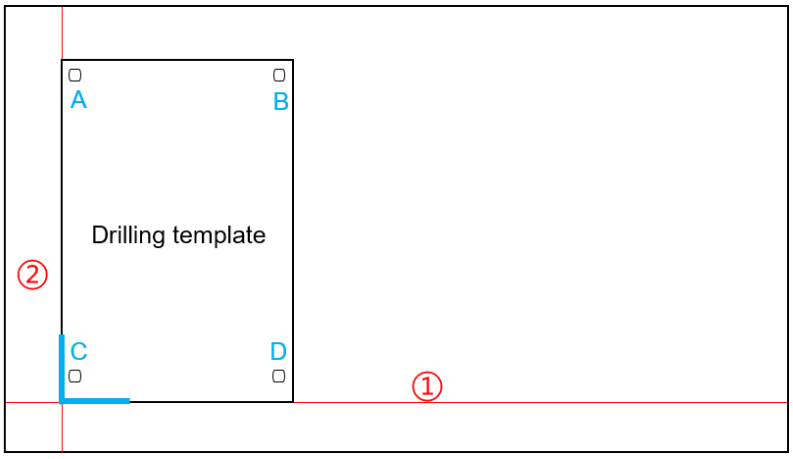


Figure 5-6: Metal Base Positioning – Marking Drilling Points

STEP 3: Drill holes vertically at the marked drilling points (hole depth: 110mm, hole diameter: 18mm). After drilling, clean out the debris from the holes.

STEP 4: Insert M16*120 expansion bolts into the holes.

STEP 5: After the first metal base is in place, align the drilling template with the first metal base to mark the anchor hole positions, and repeat the process for the remaining units.

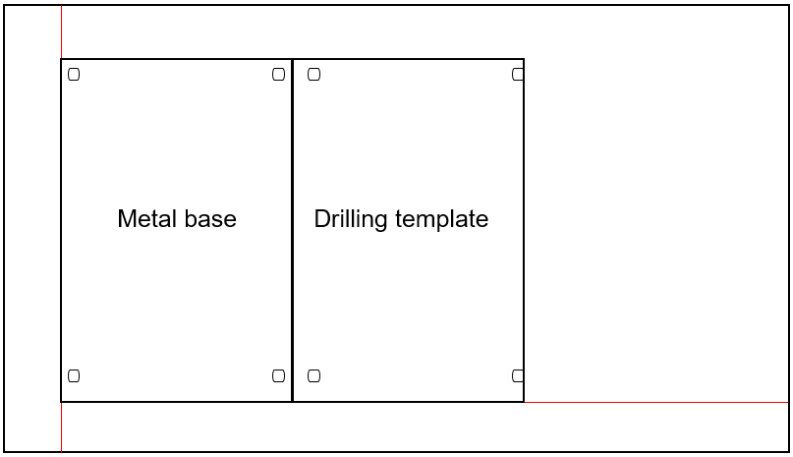


Figure 5-7: Metal Base Positioning – Aligning with Metal Base

5.2.2 Without Metal Base – Cabinet Positioning

STEP 1: Mark a reference line ① on the horizontal axis of the concrete foundation. Then, draw a reference line ② on the vertical axis on one side of the foundation, ensuring that the two lines are perpendicular to each other.

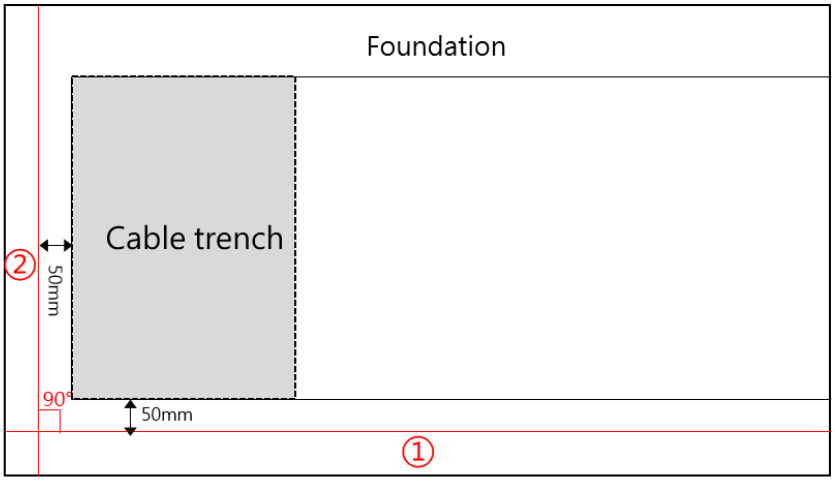


Figure 5-8: Cabinet Positioning – Drawing Lines

STEP 2: The intersection angle of the two reference lines determines the cabinet placement position (as indicated by the blue marker in the diagram below). Align one corner of the cabinet with the intersection angle and position the cabinet accordingly.

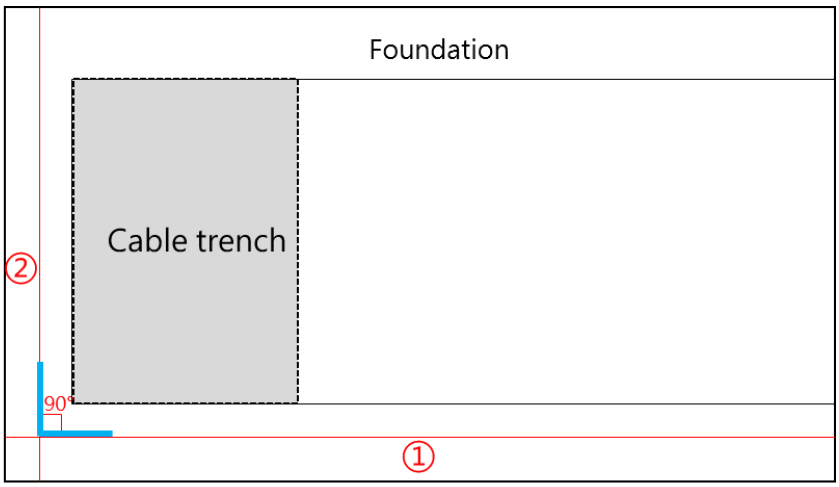


Figure 5-9: Cabinet Positioning – Marking Drilling Points

STEP 3: After the first cabinet is in place, extend a line outward from the side of the cabinet by ≥ 20 mm and mark it. Use this mark as a reference point to draw a new baseline ③ perpendicular to the original baseline ①. The intersection angle of these two lines defines the placement position for the second cabinet (as shown by the blue marker). Repeat this process for the subsequent cabinets.

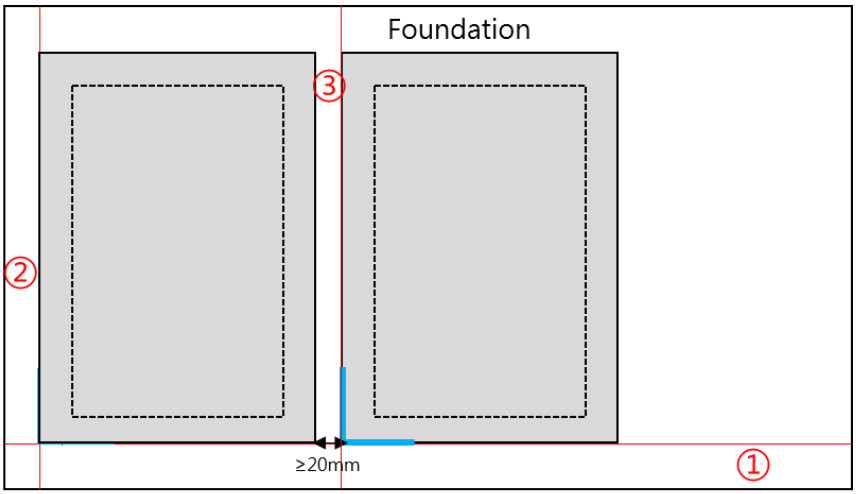


Figure 5-10: Cabinet Positioning for Multiple Cabinets

5.3 Metal Base Anchoring

STEP 1: Attach the hooks of the wire rope / lifting sling to the four holes on the base, as shown in positions ①, ②, ③, ④ in the figure below.

NOTICE

- Select appropriate wire ropes or slings according to the weight of metal base (196kg). During transportation, ensure that the center of gravity remains stable and move at a constant speed.
- The side with the cable tray partition should face the front.

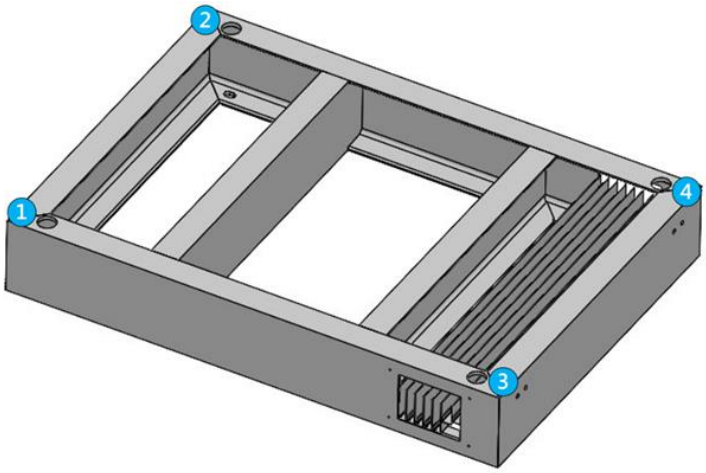


Figure 5-11: Hook Holes of Metal Base

STEP 2: Insert one of the forklift arms into the ropes and then slowly lift the metal base.

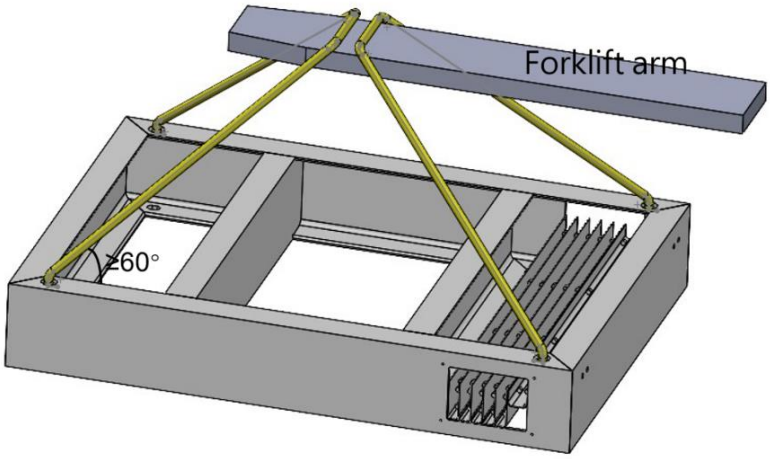


Figure 5-12: Diagram of Moving Metal Base

STEP 3: Move the metal base to the installation position. Align the four bolt holes on the base with the pre-embedded expansion bolts one by one, and then slowly lower it into place.

⚠CAUTION

If the base cannot be leveled to the ground, perform the following steps:

① Insert stainless steel shims into the gap between the base and the ground.

② Use a feeler gauge to check the gap. Stop inserting shims when the gap is ≤ 0.5 mm.

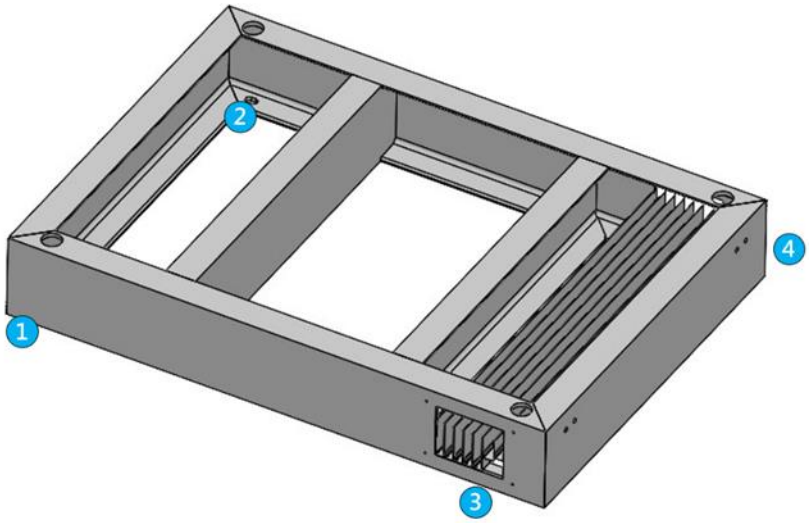


Figure 5-13: Mounting Hole Locations

STEP 4: Place the washer and nut onto the expansion bolt and tighten the nut (Torque: 160±5 N·m).

STEP 5: M10 bolts to secure the temporary positioning guides to two diagonal corners of the base. (Torque is 52 ± 5 N·m).

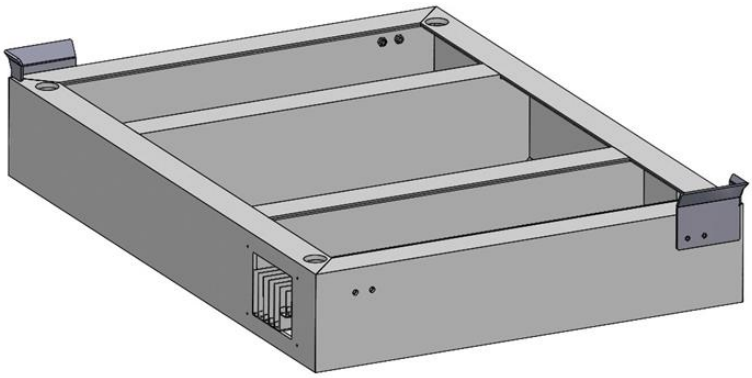


Figure 5-14: Positioning Guide Locations

5.4 Cable Routing

⚠CAUTION

- When routing power cables, ensure they are not coiled or twisted. If the cable is not long enough, it must be replaced immediately.
- When a metal base is not used, self-dug cable trenches must comply with the following requirements:
  - Cables of the same type shall be bundled together. Different types of cables (e.g., communication and power cables) must be routed separately, with a minimum spacing of 30 mm between them. Twisting or crossing of cables is strictly prohibited.
  - Labels indicating cable information must be affixed to both ends of each cable to facilitate future inspection and maintenance.

NOTICE

- For configurations with external auxiliary power, please contact Delta to obtain the auxiliary power wiring configuration document.

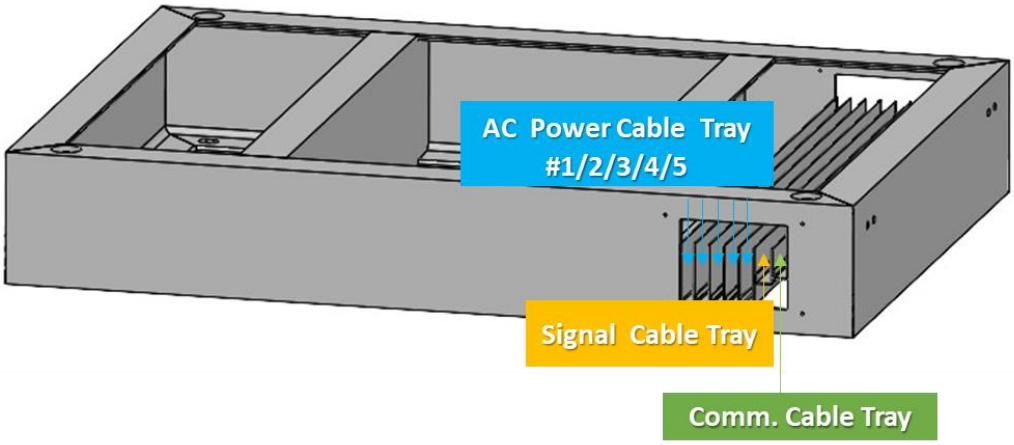


Figure 5-15: Cable Tray Layout

Standalone Cabinet

Place the following cables inside the cable tray of the metal base:

① 4 * AC power cables (R/S/T/N)

② 1 * EMS communication cable

③ Signal cables (optional, subject to project-specific requirements)

Multiple Cabinets in Parallel

① **AC power cables:** Each cabinet is equipped with four (4) AC power cables (R/S/T/N).

STEP 1: Install Cabinet #1 (refer to Section 5.5 for cabinet installation).

STEP 2: Install the junction box (refer to Section 5.7 for junction box installation).

STEP 3: Route the cables from the junction box to the corresponding cabinet bases.

Arrange the cables according to the sequence of the cable trays, as follows:

- Power cables of Cabinet #1 (L1/L2/L3/N) are placed in the first tray.
- Power cables of Cabinet #2 (L1/L2/L3/N) are placed in the second tray, and so on.

Table 5-2: AC Power Cable Laying Sequence (from Bottom to Top)

	IEC / AS	UL
Standard Version	N (Blue) → L2 (Green) → L3 (Red) → L1 (Yellow)	L1 (Yellow) → L3 (Red) → L2 (Green) → N (Blue)
Mirror Version	L3 (Red) → L1 (Yellow) → N (Blue) → L2 (Green)	L2 (Green) → N (Blue) → L1 (Yellow) → L3 (Red)

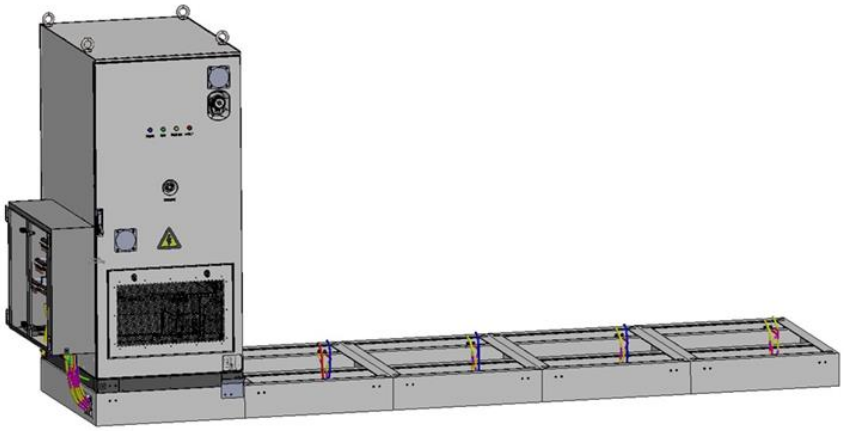


Figure 5-16: AC Power Cables Wiring

② **Inter-cabinet comm. cables:** Lay two (2) communication cables between every two (2) adjacent metal bases. For example:

- Two (2) cables between Cabinet #1 and Cabinet #2
- Two (2) cables between Cabinet #2 and Cabinet #3, and so on.

③ **Comm. cable:** Each parallel system requires one (1) EMS communication cable, which should be placed in the cable tray of the main cabinet's metal base. For communication between two sets of parallel systems (e.g., 5+5 configuration), each master cabinet shall be connected to the WAN switch via one communication cable, and the WAN switch shall then be connected to the EMS server through a single communication cable.

④ **Signal cables** (optional, subject to project-specific requirements): Lay the signal cables in the comm. cable tray of the master cabinet.

5.5 Cabinet Placement

5.5.1 Cabinet Numbering

Each cabinet is labeled with a number on the outside of the cabinet door (e.g., Cabinet #1). Cabinet 1# serves as the master cabinet and is equipped with a controller, while the remaining cabinets are designated as slave cabinets. Depending on site configuration, the cabinets can be arranged in either standard or mirrored orientation.

NOTICE

The system IP addresses are fixed and correspond to the cabinet numbering. Please install the cabinets in the specified sequence; otherwise, communication errors may occur.

Table 5-3: IP Address Corresponding to Each Cabinet

Label	BMS	PCS	EMS	IO Extender
Cabinet #1 (Master)	192.168.2.11	192.168.2.21	LAN: 192.168.2.1 WAN: 192.168.3.10	192.168.2.31
Cabinet #2 (Slave)	192.168.2.12	192.168.2.22	/	/
Cabinet #3 (Slave)	192.168.2.13	192.168.2.23	/	/
Cabinet #4 (Slave)	192.168.2.14	192.168.2.24	/	/
Cabinet #5 (Slave)	192.168.2.15	192.168.2.25	/	/

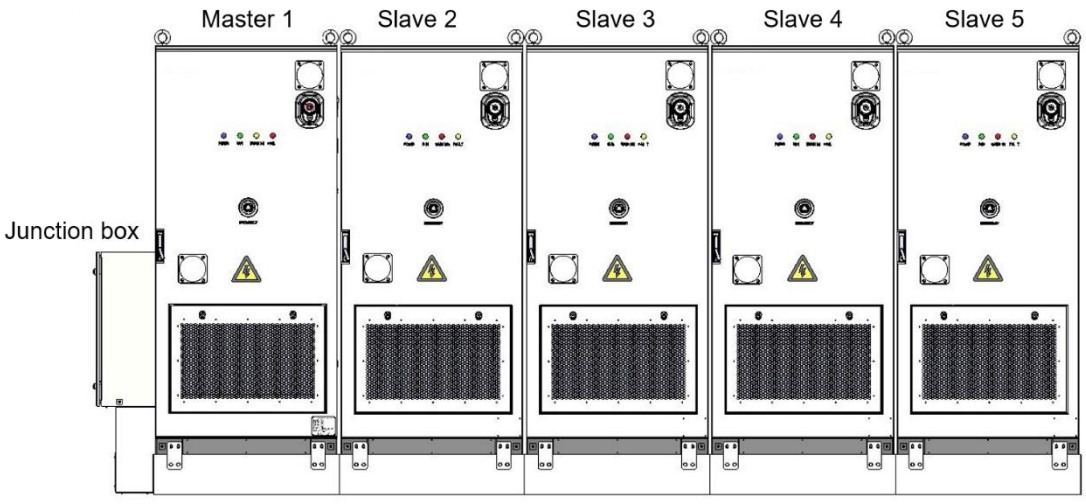


Figure 5-17: Cabinet Installation Sequence (Standard Version)

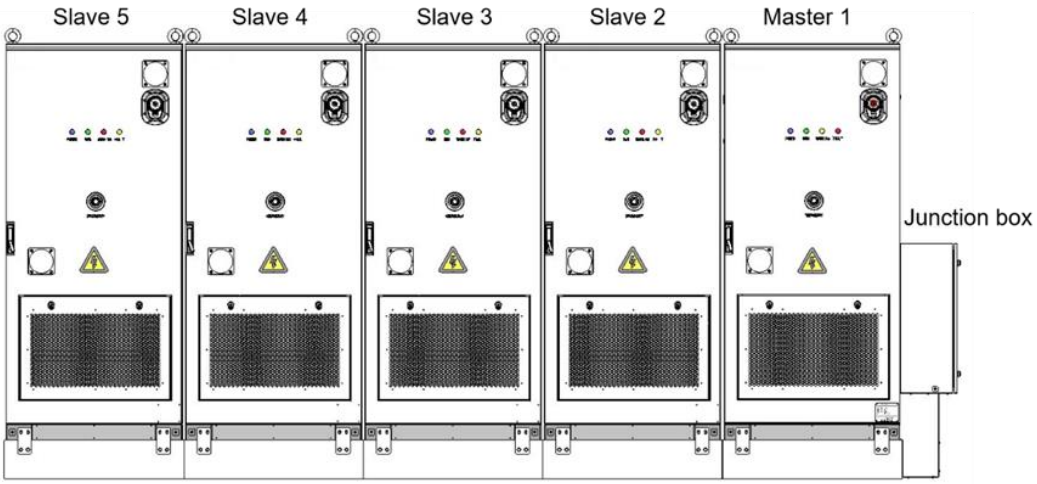


Figure 5-18: Cabinet Installation Sequence (Mirror Version)

5.5.2 Cabinet Lifting

Use a crane or forklift to move the cabinet to the installation position. Before moving, confirm the cabinet's weight and dimensions (2.7 tons, W 1000 mm × D 1420 mm × H 2250 mm). Ensure that the lifting equipment has a load capacity sufficient to handle the maximum weight of the cabinet.

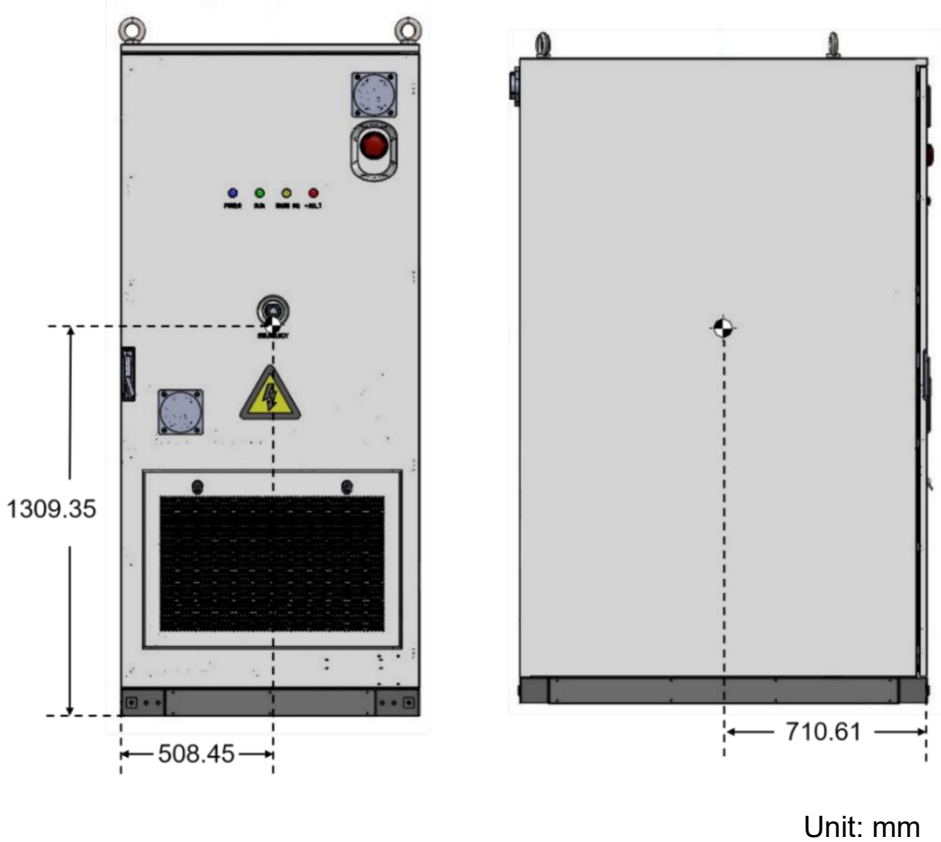


Figure 5-19: Gravity Diagram

5.5.2.1 Crane Lifting

⚠CAUTION

- Before lifting, ensure that the operators have been trained and qualified to operate the crane.
- Before lifting, ensure that the crane and slings meet the required load capacity.
- Before lifting, ensure that all cabinet doors are securely locked.
- During lifting, keep the cabinet level with the ground.
- During lifting, avoid cabinet distortion or shifting of the center of gravity

Attach the lifting slings to the lifting lugs, and secure the other ends to the crane hook. Ensure that the angle between the slings and the top of the cabinet is ≥ 60° when the slings are taut. Lift the cabinet slowly only after confirming it is level, and move it carefully to the installation position.

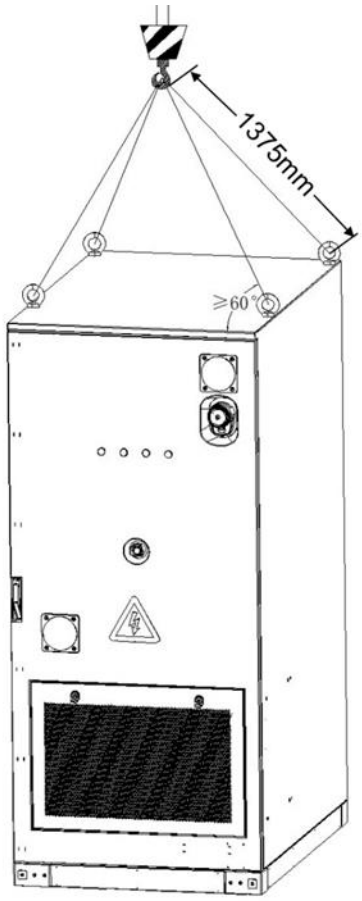


Figure 5-20: Lifting Diagram

5.5.2.2 Forklift Transport

⚠CAUTION

- Ensure that the operators have been trained and qualified to operate the crane.
- Operate on level ground. When positioning the forks, insert them into the front openings on the cabinet's base channel steel. Ensure the forks are level with the ground and that the cabinet's center of gravity is aligned with the center of the forks to prevent imbalance.
- After lifting the cabinet, drive the forklift at a safe and controlled speed. Avoid sudden movements, sharp turns, or uneven surfaces to maintain stability during transport.

STEP 1: The cabinet base is equipped with square forklift openings as shown in the figure. Insert the forklift arms through the front openings from one side to the other, then slowly lift the equipment and move it to the installation position.

① Standalone cabinet: Lift from sides (side fork pocket covers already removed)

② Multiple cabinets in parallel: Lift from front/rear (remove front/rear fork pocket cover first)

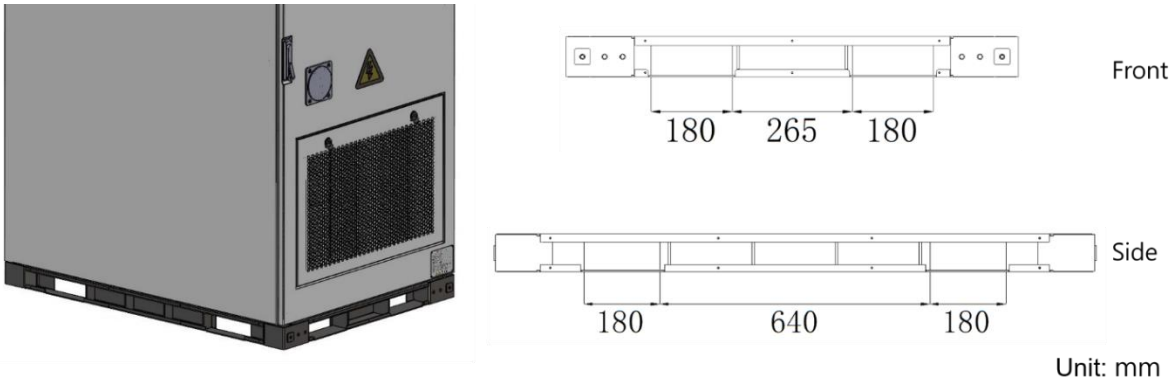


Figure 5-21: Fork Pocket Locations and Dimensions

STEP 2: Align the cabinet diagonally with the pre-installed positioning guides on the base, then slowly lower the cabinet onto the base.

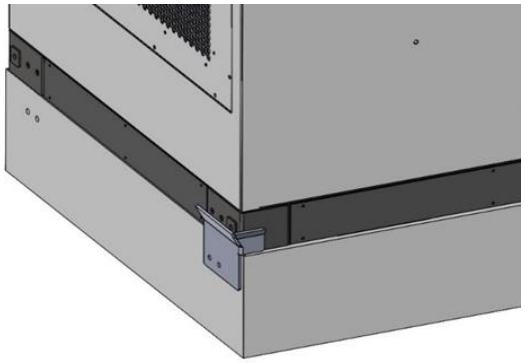


Figure 5-22: Cabinet Positioning on the Base

STEP 3: Align the fork pocket covers with the screw holes on the bottom of the cabinet. Secure it using M4 screws, tightening them in a diagonal sequence (Torque: 2 ± 0.5 N·m).

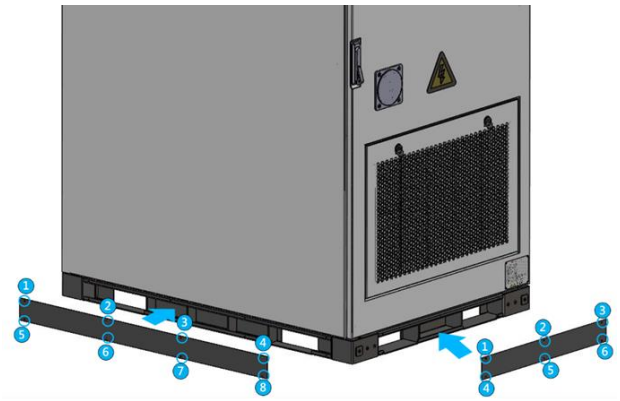


Figure 5-23: Fork Pocket Covers Installation Diagram

5.5.3 Cabinet Anchoring

The cabinet base has two fixed points at the front and rear (four in total), as shown below. Choose the appropriate fixing method according to the site conditions.



Figure 5-24: Cabinet Fixed Points

5.5.3.1 With Metal Base

STEP 1: Remove the temporary positioning fixtures pre-installed on the metal base.

STEP 2: Use four mounting brackets and sixteen M10 bolts to secure the front and rear of the cabinet to the base (Torque: 52 ± 5 N·m).

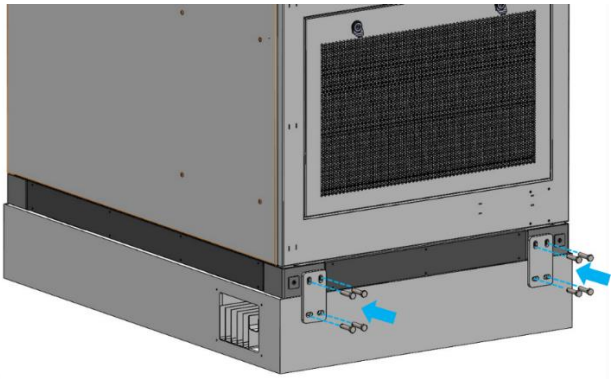


Figure 5-25: Diagram of Fixing Cabinet on the Metal Base

5.5.3.2 Without Metal Base

STEP 1: Move the cabinet to the marked installation position and keep it suspended above the ground. Once the cabinet is aligned with the markings, slowly lower it into place.

STEP 2: Align the four L-shaped brackets with the front and rear mounting holes of the cabinet, and mark the drilling points on the concrete foundation.

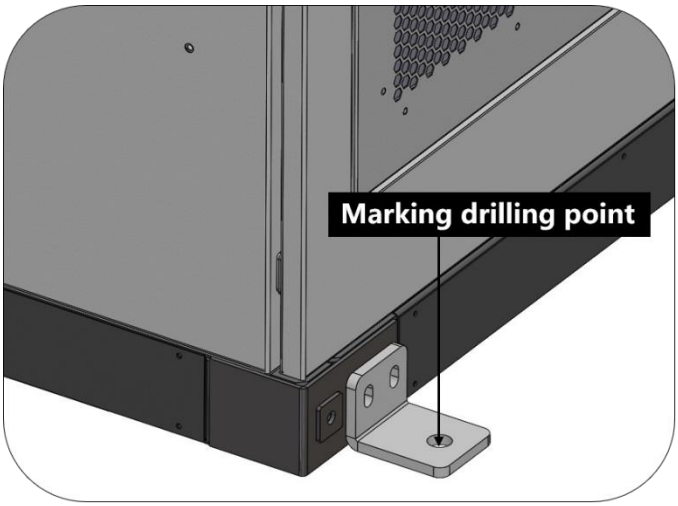


Figure 5-26: Diagram of Aligning Brackets and Marking Drilling Points

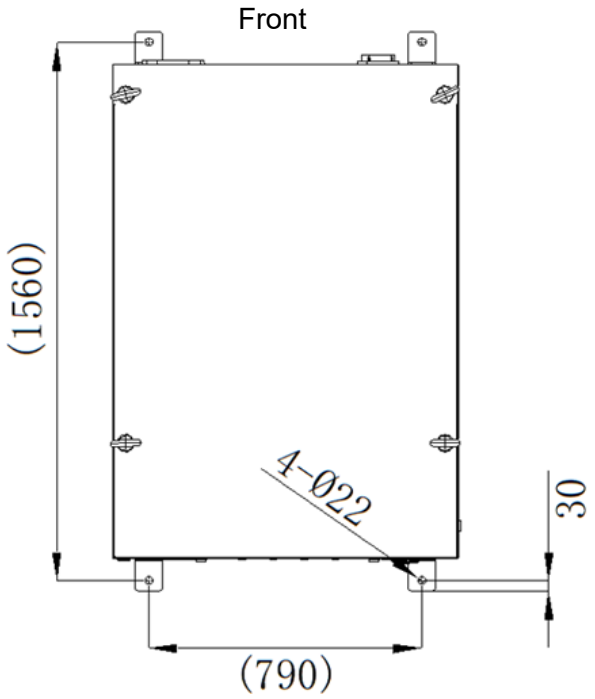


Figure 5-27: Top View of Cabinet and L-shaped Brackets

STEP 3: Remove the L-shaped brackets. Drill holes vertically at the marked drilling points (hole depth: 110mm, hole diameter: 18mm), and clean out the debris from the holes.

STEP 4: Realign the L-shaped bracket with the anchor holes, and insert M16*120 expansion bolts between the bracket and the ground. Use a hammer to drive the bolts in until they are flush with the surface, then tighten the nuts with a wrench (Torque: 160±5 N·m).

⚠CAUTION

If the L-shaped bracket cannot sit flush with the ground, perform the following steps:
① Insert stainless steel shims into the gap between the bracket and the foundation.
② Use a feeler gauge to check the gap; stop inserting shims when the gap is **≤ 0.5 mm**.
③ Fix the expansion bolts in place.

STEP 5: Secure the L-shaped brackets to the cabinet using M10 bolts (Torque: 52±5 N·m).

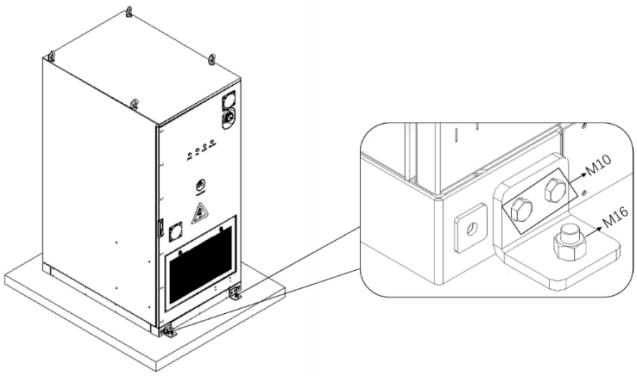


Figure 5-28: Diagram of Fixing Cabinet on the Concrete Foundation

5.6 Waterproof Plug Installation

STEP 1: Using a wrench, turn the lifting rings on the top of the cabinet counterclockwise and remove them one by one.

STEP 2: Align the waterproof plugs with the top openings and insert them vertically, then gently tap them with a rubber mallet until they are fully seated.

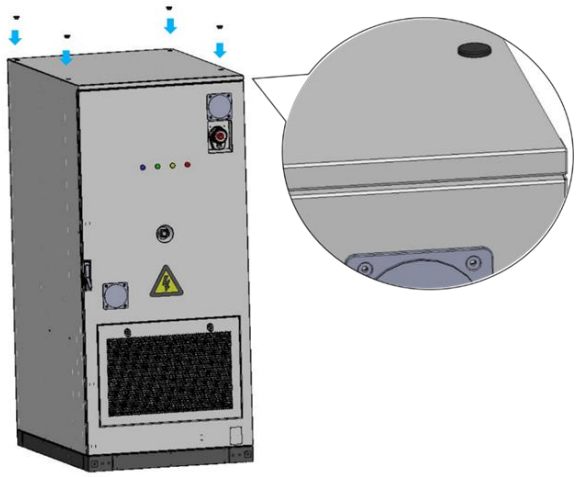
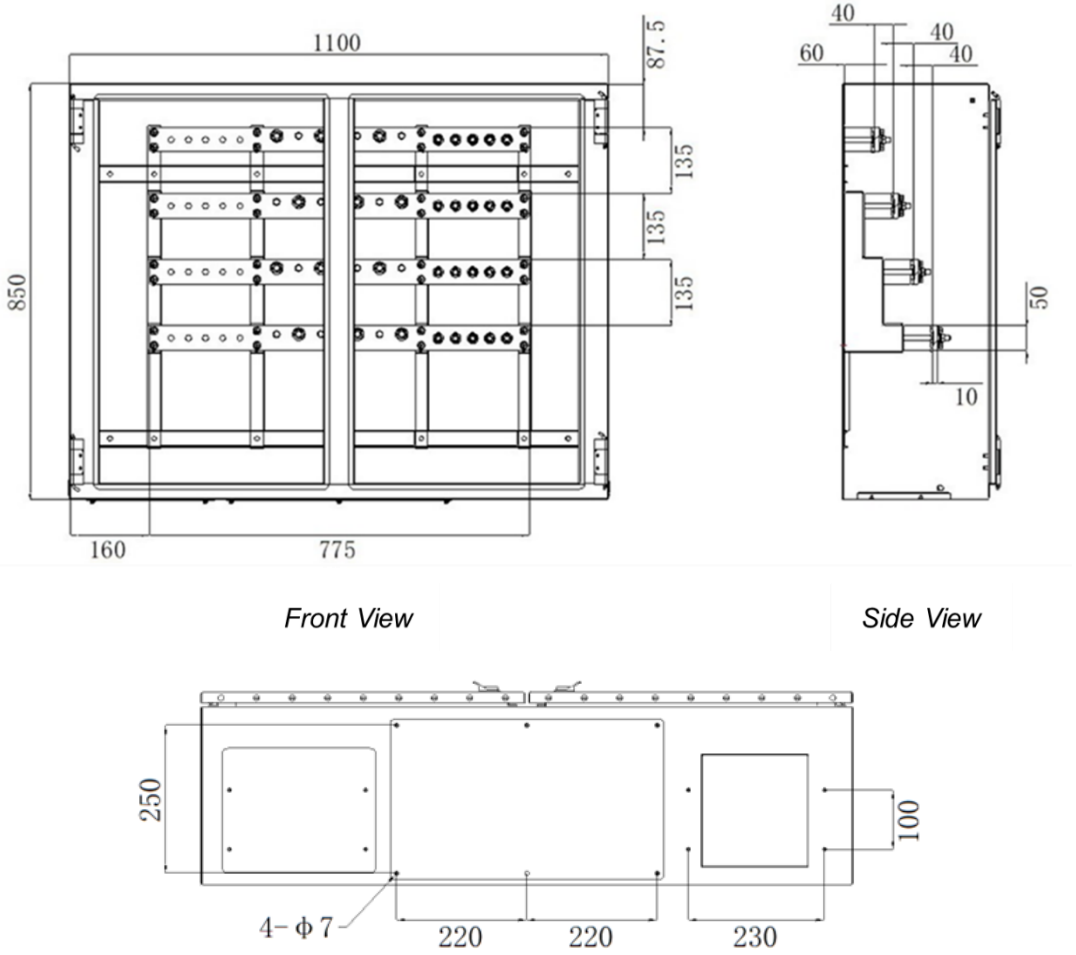


Figure 5-29: Installing Waterproof Plugs

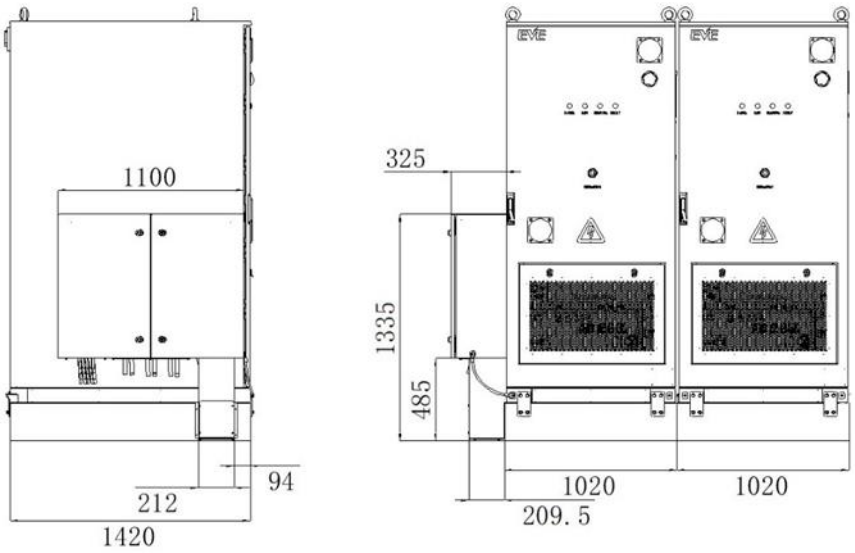
5.7 Junction Box Installation

For multi-cabinet systems, the AC power cables should be consolidated in a junction box before being connected to the external AC interface.



Unit: mm

Figure 5-30: Junction Box Dimensions



Unit: mm

Figure 5-31: Junction Box Installation Location

STEP 1: Open the door of the junction box, and align the eight (8) bolt holes inside the box with the mounting holes on the side of the main cabinet.

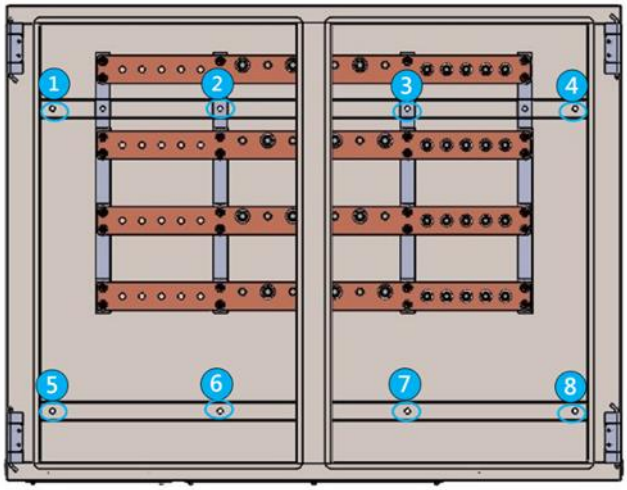


Figure 5-32: Bolt Hole Locations Inside the Junction Box

STEP 2: Secure the junction box using eight (8) M6 bolts (Torque: 8±1 N·m).

5.8 Cable Installation

5.8.1 Grounding Cables

⚠WARNING

- Connect the grounding cable before connecting the AC and DC wiring.
- The grounding connections must comply with the grounding standards and regulations of the country or region where the project is located.
- The grounding grid and grounding cables shall be installed during the foundation construction stage.
- According to IEC grounding standards, this product is compatible with both TN-S and TT-N grounding configurations, and the corresponding grounding connections and protective arrangements shall be implemented as required by each architecture; if a PEN system is used, an isolation transformer shall be installed.

① TN-S:

- The system grounding shall not share the same grounding conductor with other equipment to minimize interference.



Figure 5-33: TN-S Grounding Architecture

② TT-N:

- A Residual Current Device (RCD) shall be installed on the main energy-storage circuit to provide the required fault protection.

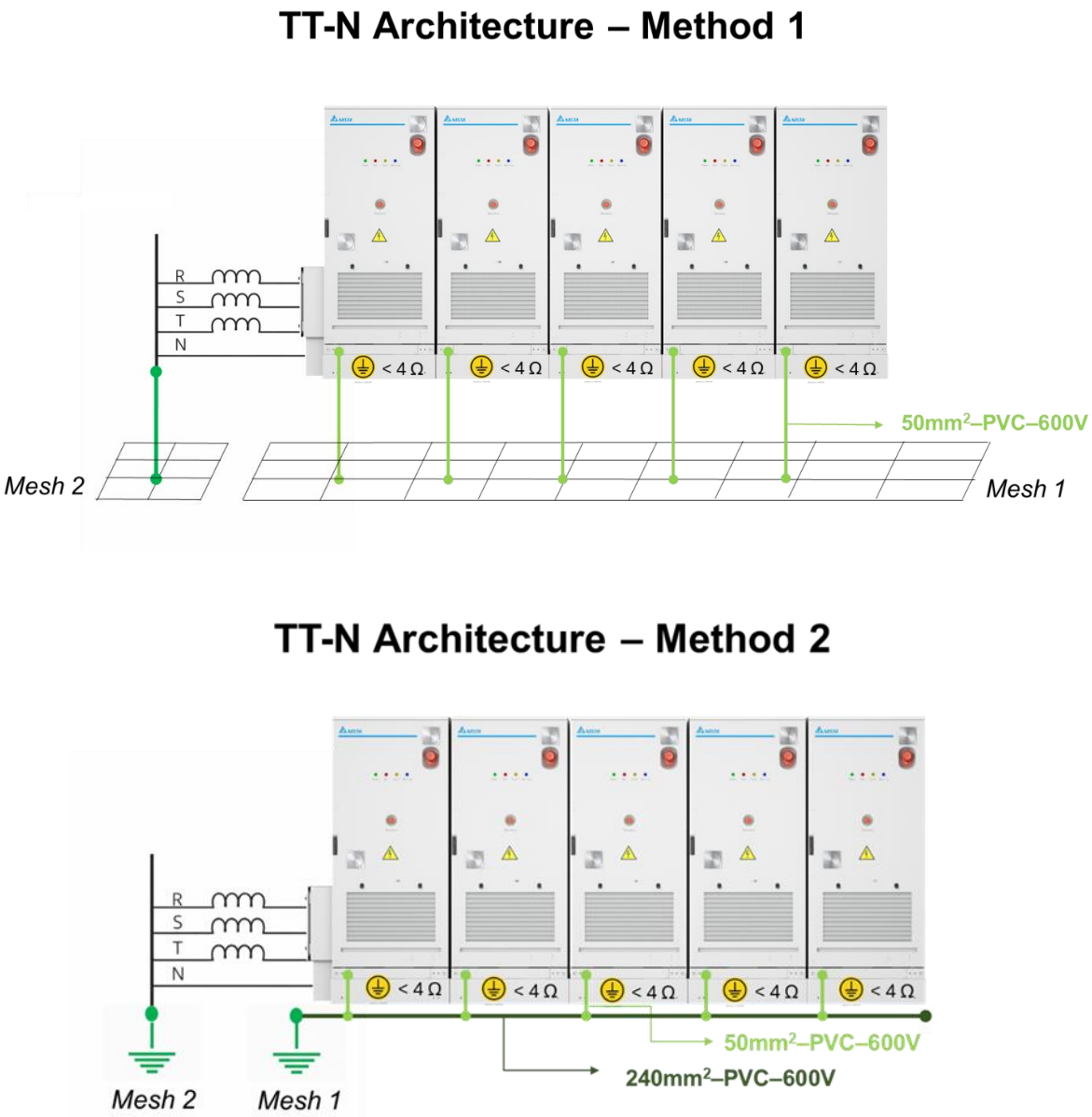


Figure 5-34: TT-N Grounding Architecture

STEP1: Connect the following three types of grounding cables.

① Junction box grounding: use the grounding cable and M8 bolts provided with the junction box to connect the junction box grounding terminal to the main cabinet grounding terminal. Tighten to the specified torque (25 ± 1 N·m).

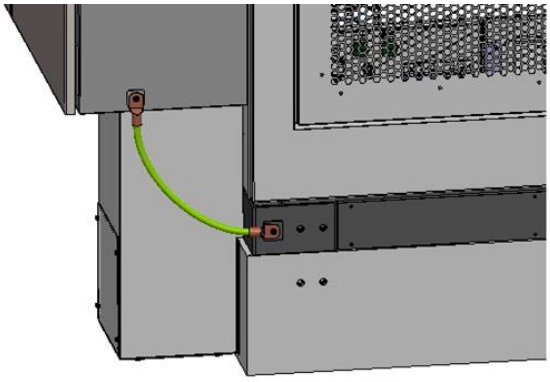
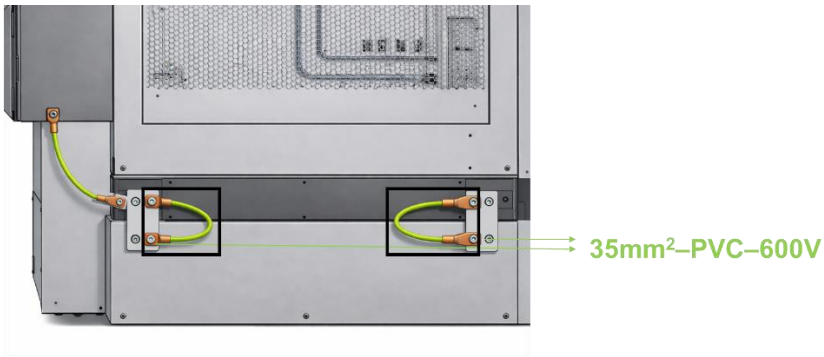


Figure 5-35: Junction Box Grounding Connection

② Metal base grounding: use the M10 bolts and mounting brackets provided in the metal base kit, together with the user-prepared grounding cable (spec. as follows), to bond the base of each cabinet to the metal base. Tighten to the specified torque (42 ± 5 N·m).



Method	Spec.	QTY	Scope
1	50mm²-PVC-600V	1	User-prepared
2	35mm²-PVC-600V	2	

Figure 5-36: Metal Base Grounding Connection

③ System grounding: ground in accordance with the TN-S or TT-N earthing system configuration.

NOTICE Each cabinet has two grounding terminals at the front and rear.



Figure 5-37: Grounding Points

STEP 2: After grounding is completed, test the grounding resistance. The resistance value should be less than 4Ω.

5.8.2 Power and Comm. Cables

⚠ DANGER

- Before performing any electrical connections, ensure that all switches in the cabinet are in the OFF state. Failure to do so may result in high-voltage electric shock hazards.
- Before touching any conductive surfaces or terminals, measure the voltage at the contact point and ensure the equipment is properly grounded.

5.8.2.1 Standalone Cabinet

1. AC Power Cable

STEP 1: Remove the protective cover on the MCCB and the cable entry plate.

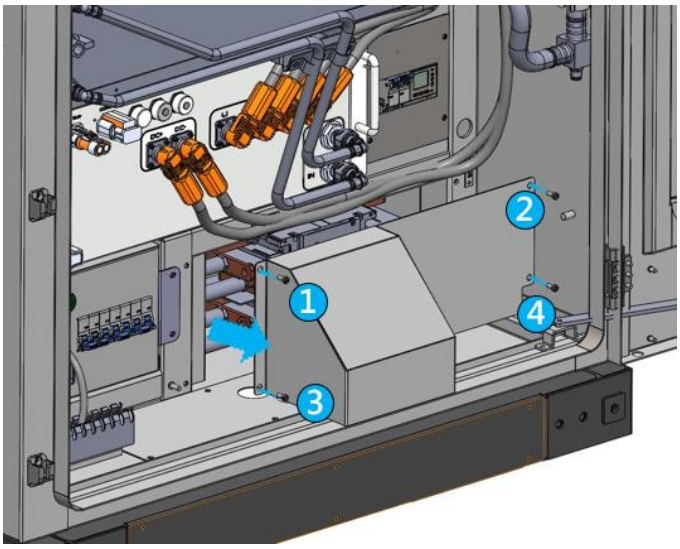


Figure 5-38: MCCB Protective Cover Removal

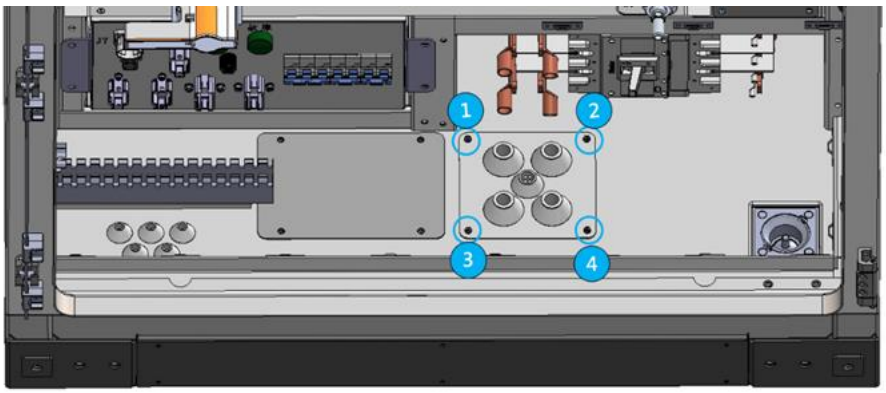


Figure 5-39: Cable Entry Plate Removal

STEP 2: Route four (4) power cables through the four (4) cable entry holes from the bottom of the plate, and guide them to the MCCB.

STEP 3: Secure one end of the four (4) cables to the L1/L2/L3/N terminals on the MCCB using four (4) M10 bolts (Torque: 42 ± 5 N·m), and the other end to the external AC side.

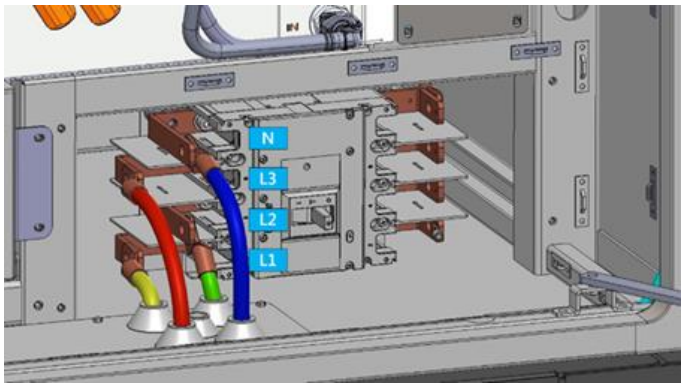


Figure 5-40: AC Power Cable Connection Terminals (IEC / AS Version)

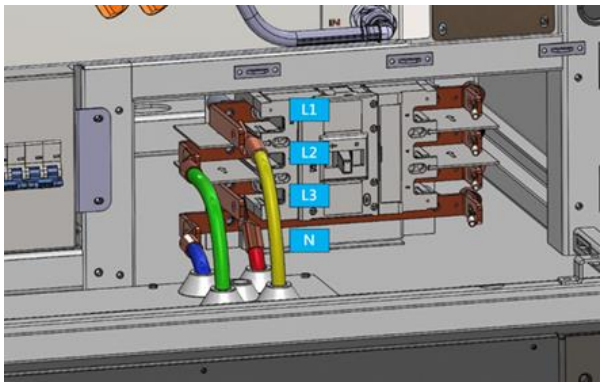


Figure 5-41: AC Power Cable Connection Terminals (UL Version)

STEP 4: Reinstall the cable entry plate using four (4) M5 bolts in its original position. Then, secure the protective cover with four (4) M5 bolts. (Torque: 5 ± 0.5 N·m)

2. EMS Comm. Cable

Remove the cable entry plate, route the EMS communication cable through the cable entry hole, and connect it to the EMS communication interface. The other end of the cable shall be connected to the external EMS server.

NOTICE Signal cables (optional) can also be routed into the cabinet in this step.

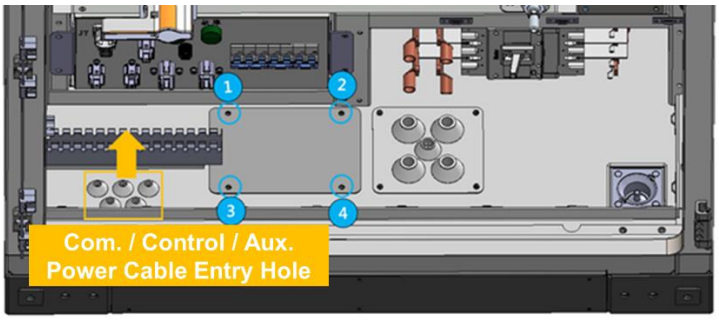


Figure 5-1: Comm. Cable Entry Plate and Entry Hole

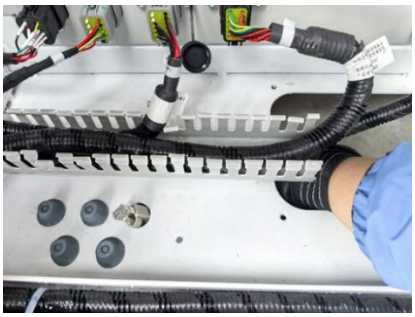


Figure 5-2: Comm. Cable Routing

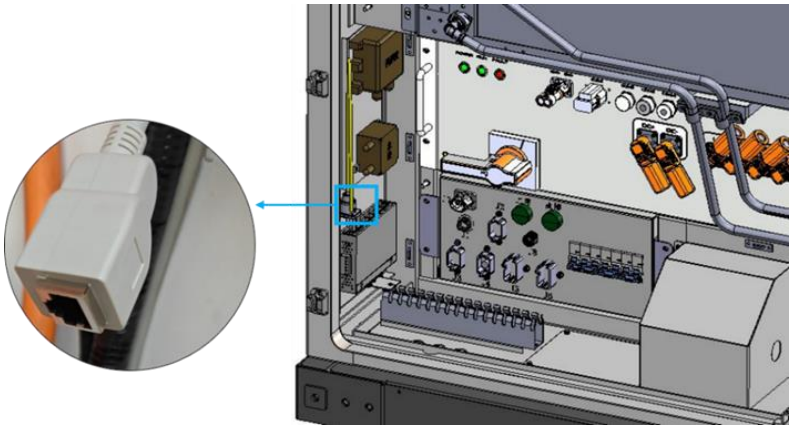
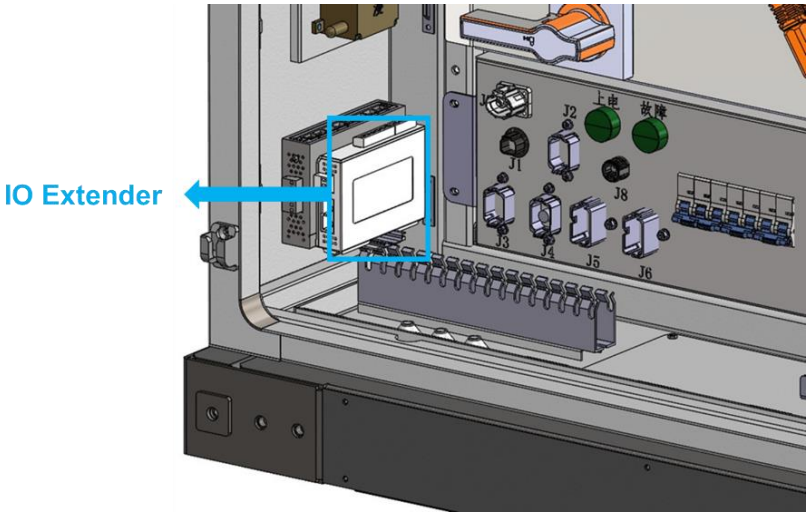
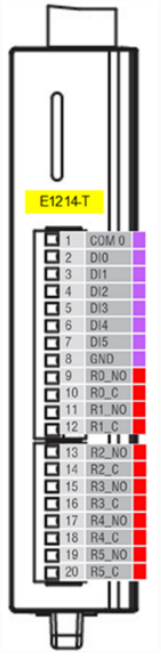


Figure 5-42: EMS Communication Interface

4. Signal Cable (Optional)

The signal cables shall be installed on the I/O extender of the master cabinet. The port definitions are as follows:





DI/DO	Function	Normal Operation	Fire Alarm Triggered	LV Shutdown
DI0	Site Access			
DI1	Site EPO			
DI2	Site Fire Alarm -1			
DI3	Site Fire Alarm - 2			
DI4	Reserved			
DI5	Reserved			
R0	Level 2 Fire Alarm triggered in any cabinet	NC	Open	Open
R1	#1 Cabinet Level 1 Fire Alarm	NC	Open	Open
R2	#2 Cabinet Level 1 Fire Alarm	NC	Open	Open
R3	#3 Cabinet Level 1 Fire Alarm	NC	Open	Open
R4	#4 Cabinet Level 1 Fire Alarm	NC	Open	Open
R5	#5 Cabinet Level 1 Fire Alarm	NC	Open	Open

Figure 5-43: I/O Extender Location and Port Definition

5.8.2.2 Multiple Cabinets in Parallel

DELTerra C supports up to two grid-connected systems operating in parallel. Each system can connect up to five cabinets in parallel.

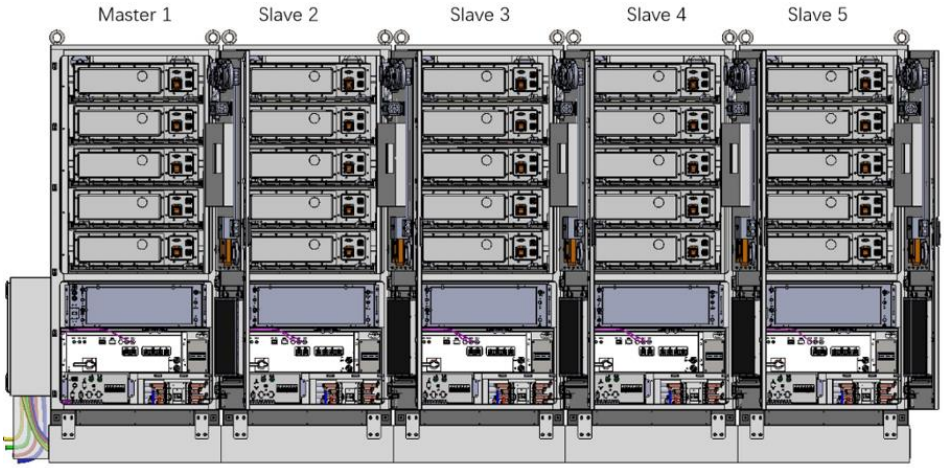


Figure 5-44: Master/Slave Configuration Diagram

1. AC Power Cables of Cabinets

STEP 1: Remove the MCCB protective cover and cable entry plate of each cabinet. Then, connect the AC power cables of each cabinet to the L1/L2/L3/N terminals of the MCCB in sequence (refer to Section 5.8.2.1 for specific steps).

STEP 2: Connect the other ends of the cables to the L1/L2/L3/N terminals in the junction box using M10 bolts (Torque: 42±5 N·m).

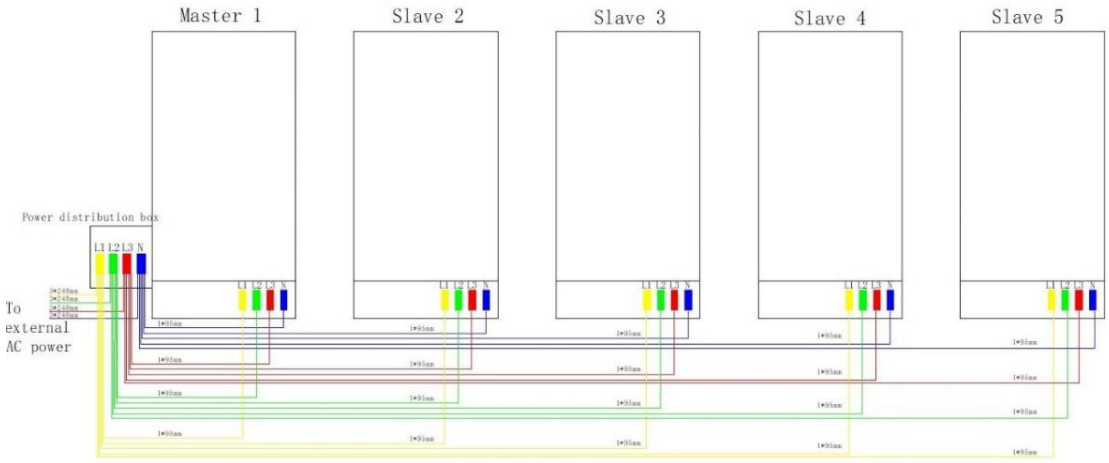


Figure 5-45: AC Power Cable Connection Diagram

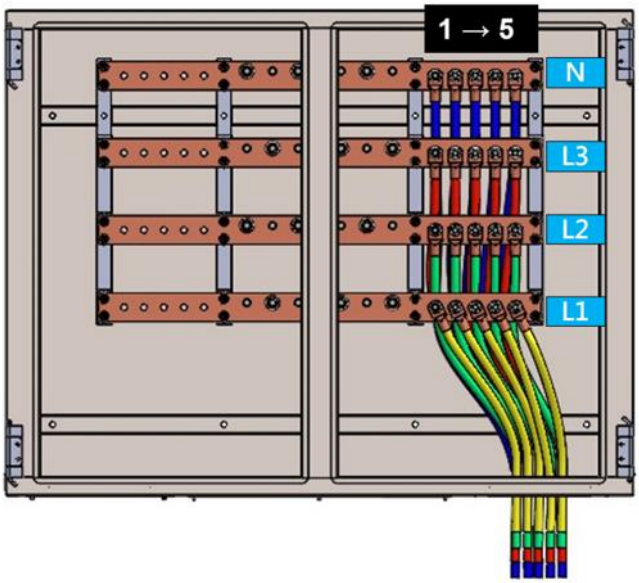


Figure 5-46: AC Power Cable (From Cabinets) Connection Terminals in the Junction Box (Standard)

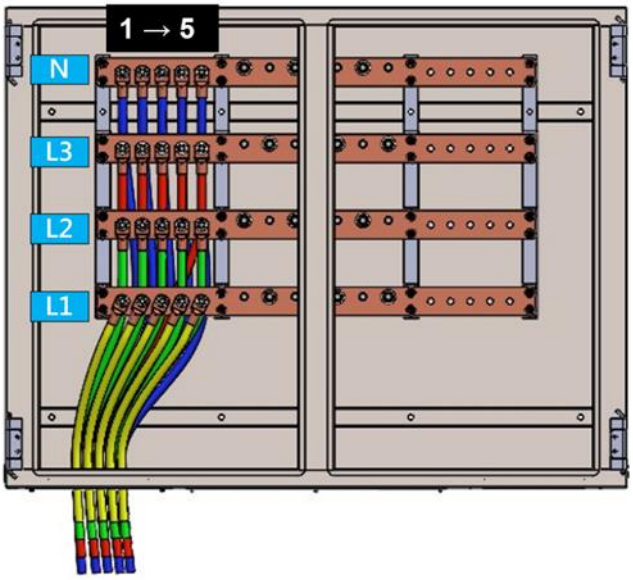


Figure 5-47: AC Power Cable (From Cabinets) Connection Terminals in the Junction Box (Mirror)

2. Main AC Power Cables

Connect the AC power cables to the L1/L2/L3/N busbars located at the center of the junction box. Up to three sets (a total of 12 cables) can be installed. To ensure proper layout and heat dissipation, it is recommended to follow the staggered routing method shown in the figure. After positioning the cables, secure them using the M12 screws provided in the cabinet (Torque: 90 ± 10 N·m).

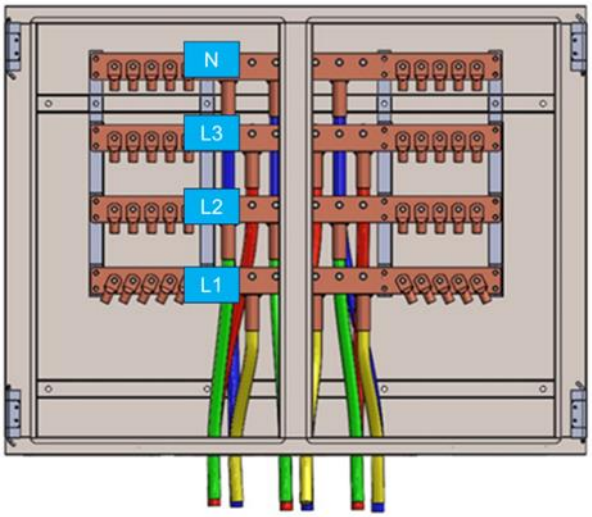


Figure 5-48: AC Power Cable (To Distribution Panel) Connection Terminals in the Junction Box

3. Communication Cables

STEP 1: Remove the cable entry plate, route the pre-laid communication cables through the cable entry hole (refer to Section 5.8.2.1 for specific steps).

NOTICE Signal cables (optional) can also be routed into the cabinet in this step.

STEP 2: Daisy-chain the switches and PCS communication interfaces of each unit, and connect the EMS communication interface of the master unit to the EMS server (for a single-system configuration: 1~5 cabinets) or to the WAN switch (for a dual-system configuration: 6~10 cabinets).

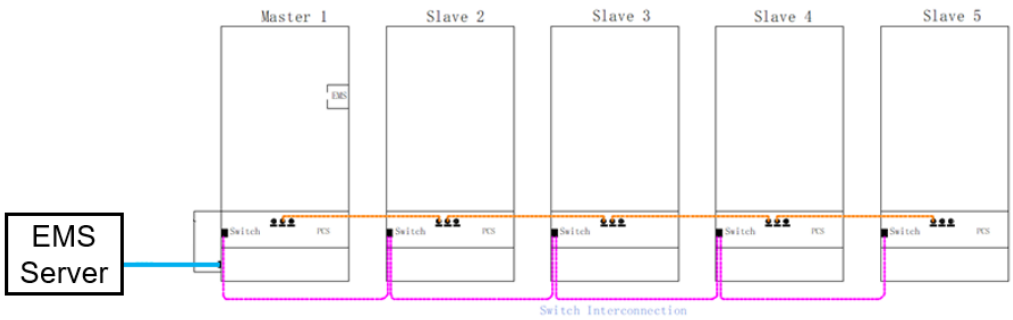


Figure 5-49: Single-system Communication Cable Connection Diagram

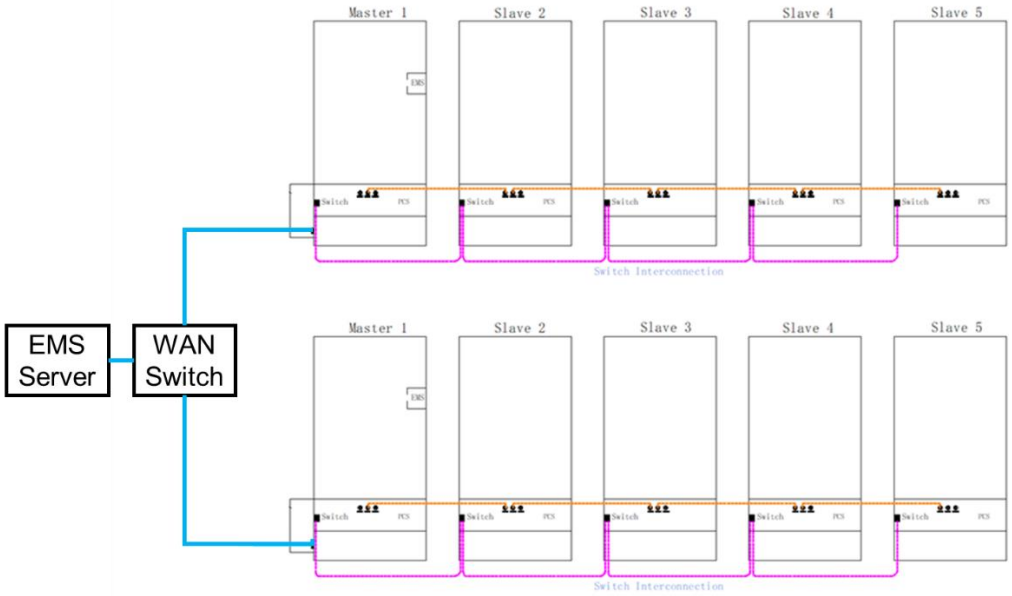


Figure 5-50: Dual-System Parallel Communication Cable Connection Diagram

3.1 PCS Interconnection

STEP 1: Daisy-chain the COM 2 and COM 3 ports between adjacent PCS units. For example, connect the COM3 port of Cabinet #1 to the COM2 port of Cabinet #2, then connect the COM3 port of Cabinet #2 to the COM2 port of the next cabinet, and so on.

STEP 2: A PCS comm. terminator must be inserted into the master unit's (Cabinet #1) COM2 port and the last slave unit's COM3 port.

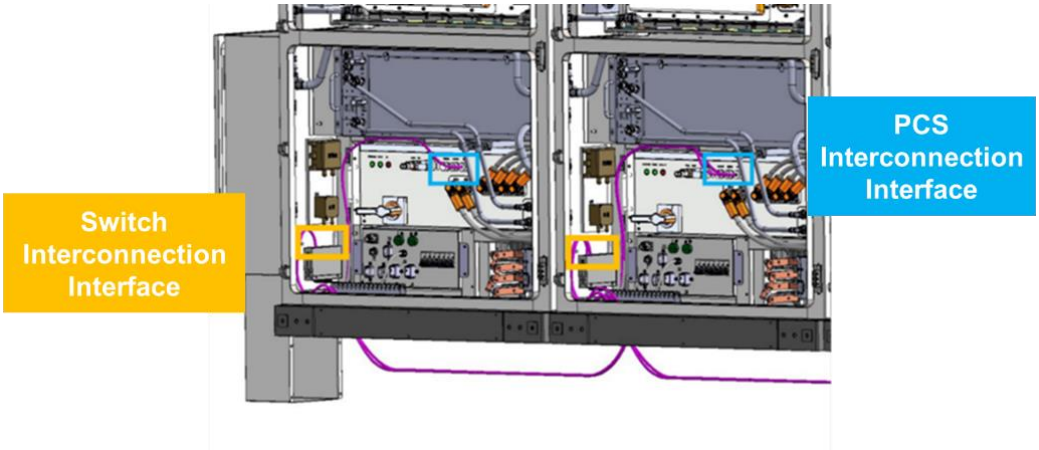


Figure 5-51: PCS Daisy-Chain Wiring Diagram



Figure 5-52: PCS Communication Interface

3.2 Switch Interconnection

Daisy-chain the switches inside each cabinet. For example, connect an available port on the switch in Cabinet #1 to any available port on the switch in Cabinet #2, then connect Cabinet #2 to Cabinet #3, and so on.



Figure 5-53: Switch Location

3.3 EMS Interface Connection

Connect one end of the EMS communication cable to the reserved interface shown in the figure below, and connect the other end via one communication cable to the EMS server or the WAN switch.

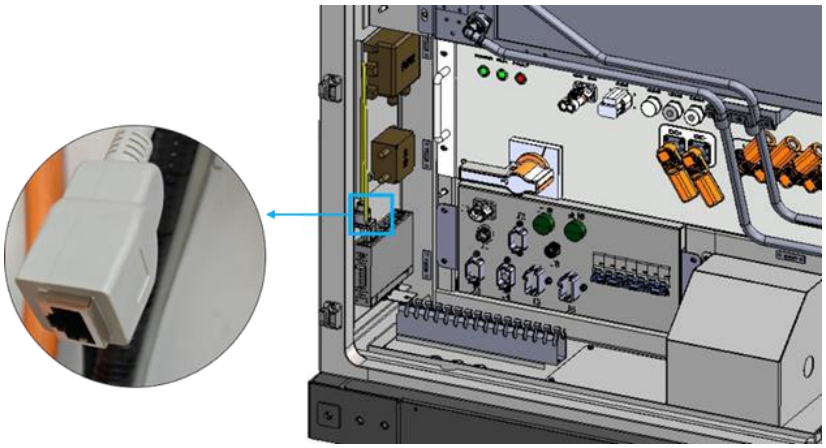


Figure 5-54: EMS Communication Interface

5. Signal Cables (Optional)

Please Refer to Section 5.8.2.1 for the port assignment.

5.9 Cover Plate Installation

⚠CAUTION

- Any openings on the internal or external cover plates of the cabinet and the junction box shall be properly sealed to prevent the ingress of animals and moisture into the cabinet.
- All external cables connected to DELTerra C shall be provided with appropriate protective measures (such as conduits, cable trays, or sheaths) to prevent damage caused by mechanical impact, environmental factors, or external interference.

Standalone Cabinet

- **Cable Entry Side** (① in the diagram below): Pass the following cables through the cover plate, which shall have pre-drilled cable holes prepared by the user. Then, secure the plate with four (4) M5 bolts (Torque: 5 ± 0.5 N·m)
  1. Four (4) AC power cables
  2. One (1) EMS comm. cable
  3. Signal cables (optional, depending on project requirements)
- **Rear Side** (② in the diagram below): Use four (4) M5 bolts to secure the cover plate (Torque: 5 ± 0.5 N·m).

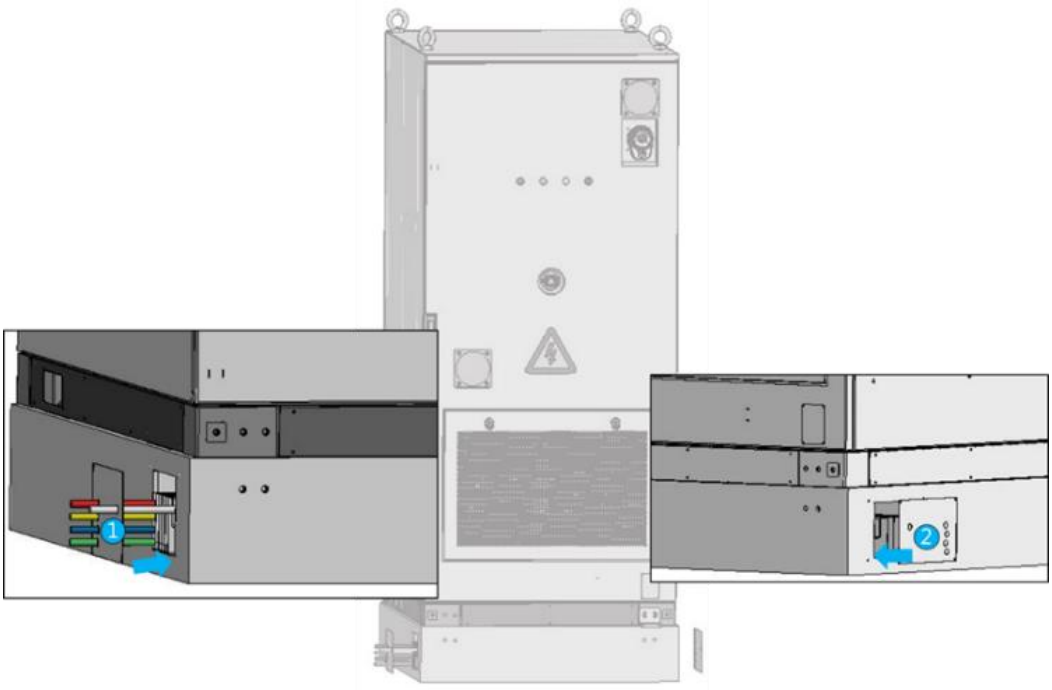


Figure 5-55: Cover Plate Locations – Standalone Cabinet

Multiple Cabinets in Parallel

- **Cable Entry Side** (① in the diagram below): Pass the following cables through the cover plate, which shall have pre-drilled cable holes prepared by the user. Use four (4) M5 bolts to secure both sides of the plate to the cable entry, then use another four (4) M5 bolts to fasten the top of the plate to the junction box (Torque: 5 ± 0.5 N·m).
  1. One (1) EMS communication cable
  2. Signal cables (optional, depending on project requirements)
- **Rear Side** (② in the diagram below): Use four (4) M5 bolts to secure the cover plate (Torque: 5 ± 0.5 N·m).

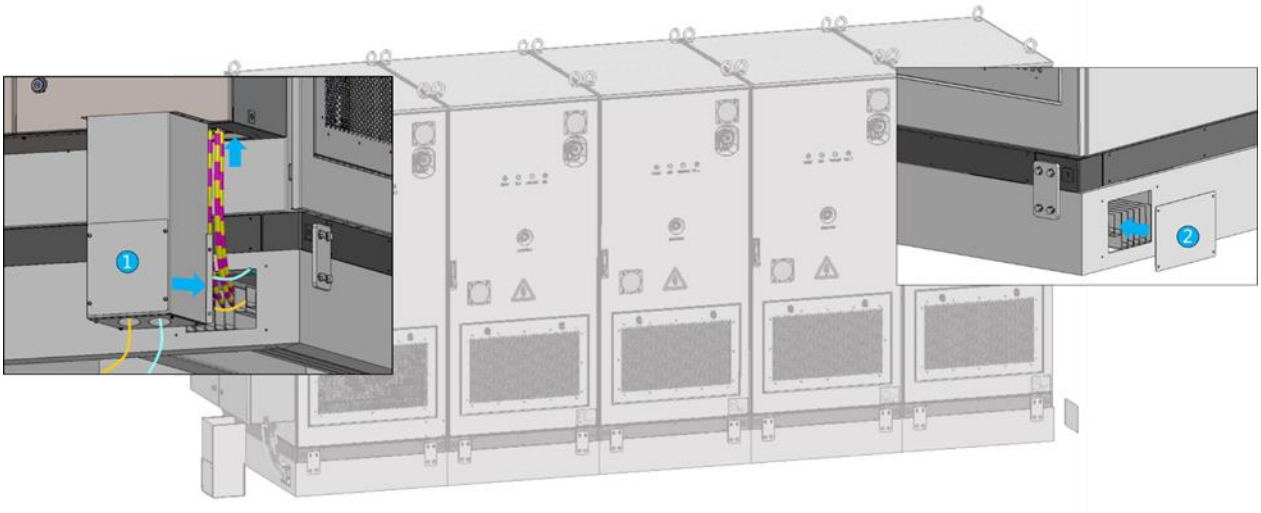


Figure 5-56: Cover Plate Locations – Multiple Cabinets in Parallel

5.10 MSD Installation

The Manual Service Disconnect (MSD) is designed to isolate the battery during maintenance, preventing remote activation of the battery rack and thereby avoiding potential injury to on-site personnel.2

DANGER The MSD shall not be switched under load. Always isolate and de-energize the circuit before operating the MSD.

STEP 1: Press the PRESS button on the MSD.
STEP 2: Open the black handle.
STEP 3: Insert the MSD into the corresponding slot on the battery module.
STEP 4: Push the black handle upward and press the red key to lock the MSD.

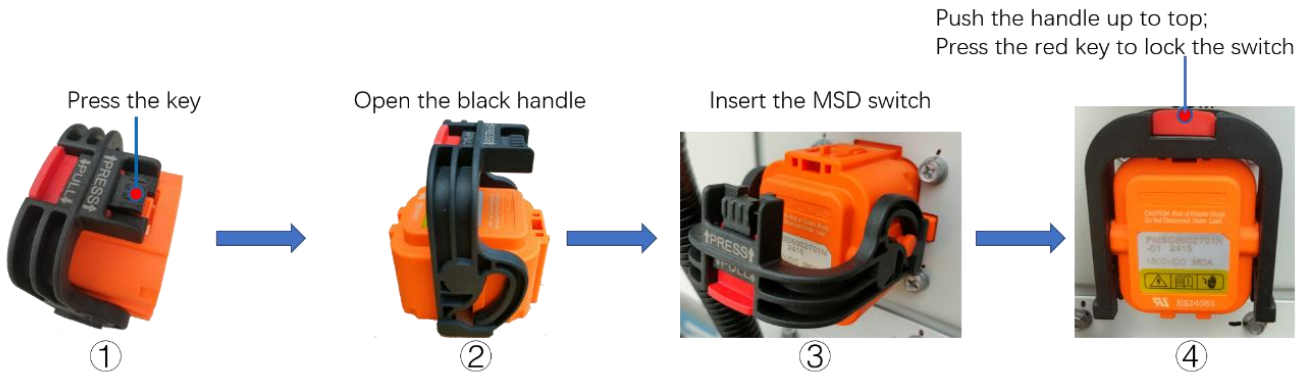


Figure 5-57: MSD Installation Process



Figure 5-58: MSD Installation Location

5.11 Post-installation Check

Table 5-4: Check List - Appearance

No.	Checklist
1	No paint peeling, damage, or deformation on the equipment exterior.
2	Equipment is securely fastened to the base.
3	Installation space meets maintenance requirements.
4	Cable connections are consistent with the manual and are securely connected.
5	Torque marking (a straight line) is applied at the junction of fastened components.
6	All MSDs are installed and locked.
7	Installation area is clean and free of tools or foreign objects.
8	All protective covers and insulating sleeves for electrical components are in place.

Table 5-5: Check List – Testing

No.	Checklist
1	The grounding resistance < 4Ω.
2	Check the torque of each power and grounding cable connection to ensure it meets the required specification.

5.12 Fire Protection System Setup

⚠CAUTION

- The cabinet is equipped with a water inlet. Whether a firefighting water injection system is required shall be evaluated and planned based on site-specific conditions and local fire regulations.
- If a firefighting water injection system is to be installed, a corresponding drainage system must also be constructed. The drainage outlet shall include an interception mechanism to ensure effective containment and collection of firefighting wastewater upon fire alarm activation; prior to installation, verify that the fire hose fitting and equipment-side water interface are of compatible types, and ensure the hydrant valve is in the closed position.

NOTICE

- During water injection, continuous water leakage may occur from the battery compartment floor drain (Ø19 mm) and the gas-venting valve (Ø92 mm).

Firefighting Water System Design Reference

- Specification of firefighting water inlet: DN65
  - IEC/AS Version: G2 1/2" female thread (ISO 228-1)
  - UL Version: 2.5" female NPT thread (ASME B1.20.1)
- Recommended water pressure: 20kPa
- Recommended flow rate: 21 L/min

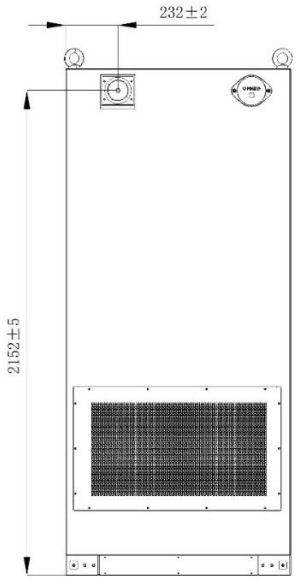


Figure 5-59: Water Inlet Position (Rear View)

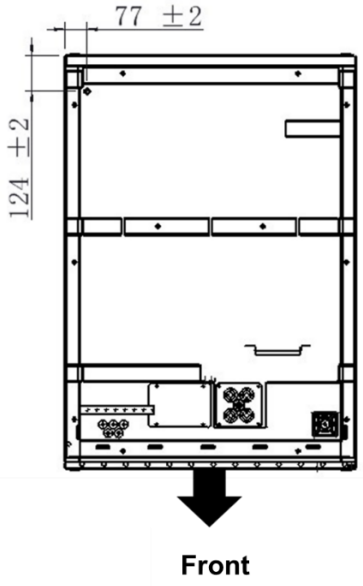


Figure 5-60: Floor Drain Position (Top View)

Fire Extinguishing Device Connection

The fire extinguishing device is shipped in a disconnected state. After all cabinets have been fully installed and commissioned, connect the wiring-harness conduit by turning it clockwise onto the interface located on the left side of the fire extinguishing device. The device can then be put into operation.

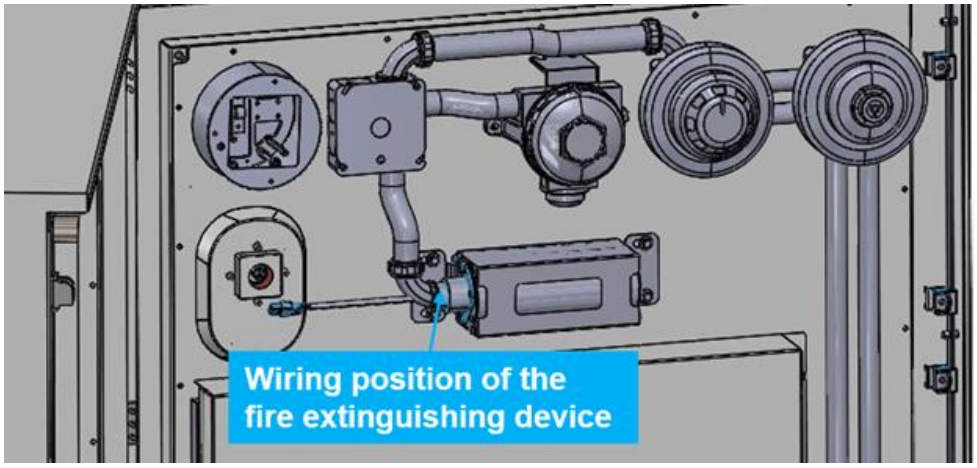


Figure 5-61: Wiring Position of the Fire Extinguishing Device

6 OPERATION GUIDE

6.1 Power-on Operation

6.1.1 Inspection

STEP 1: Confirm all the breakers are in the OFF position.

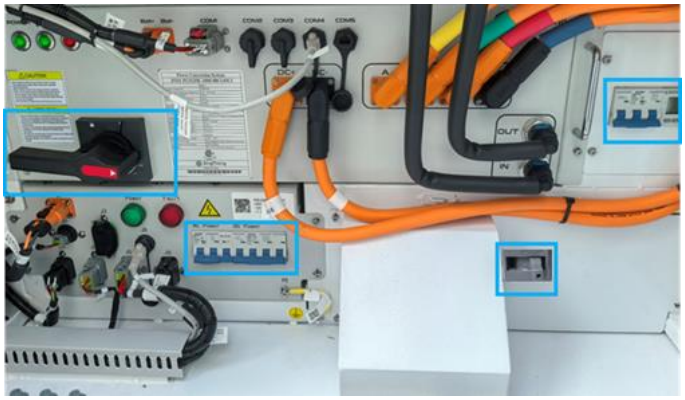


Figure 6-1: Breakers in OFF Position

STEP 2: Check that all the power cables are securely connected.

STEP 3: Check that the connection terminals of all communication and power cables are securely connected.

STEP 4: Ensure the E-stop button on the cabinet door is in its normal (released) position.

6.1.2 System Power-On

STEP 1: Switch the MCCB (QF0) to the ON position.

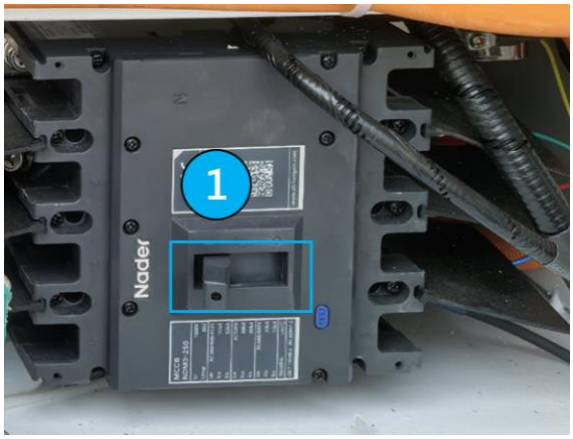


Figure 6-2: MCCB (QF0) in the ON Position

STEP 2: Turn on the liquid cooling power breaker.

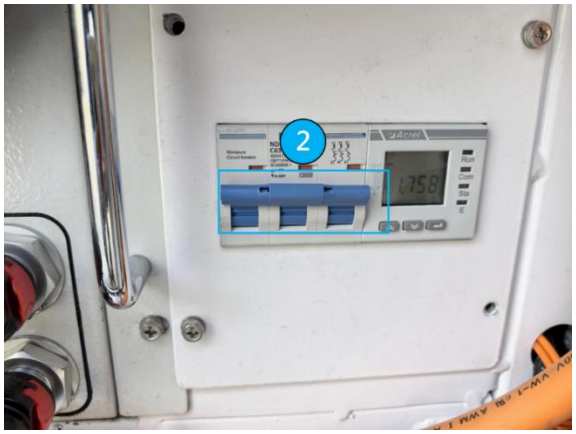


Figure 6-3: Liquid Cooling Power Breaker in the ON Position

STEP 3: Turn the PCS isolation switch to the ON position (clockwise).



Figure 6-4: PCS Isolation Switch in the ON Position

STEP 4: Turn on the AC and DC breakers on the control box. The Power indicator will illuminate.



Figure 6-5: AC and DC Breakers in the On Position

NOTICE

- If only the AC breaker is closed, the system is powered by the external AC supply.
- If only the DC breaker is closed, the system operates on internal battery power.
- When both AC and DC breakers are closed, the system normally operates on the external AC power. In the event of an external power failure, the system automatically switches to internal battery power supply to ensure uninterrupted operation.

After completing the above steps, if the system self-check indicates no equipment faults, charging and discharging operations can be performed by sending power commands through the EMS.

6.2 Indicator Light Introduction

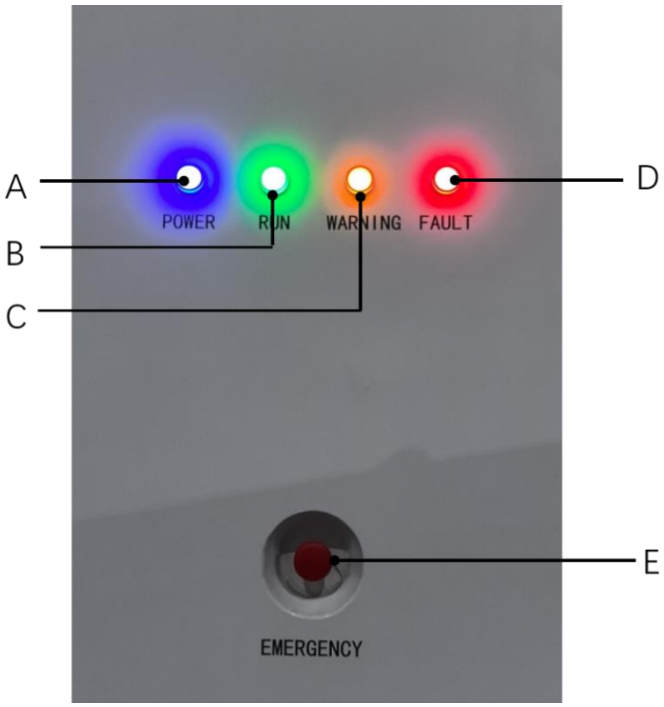


Figure 6-6: Indicator Light and EPO Layout

Table 6-1: Indicator light/button introduction

NO.	Light/Button	Description
A	POWER	The indicator light turns on when auxiliary power is supplied, either from an external source or from the battery.
B	RUN	The indicator light turns on when the system is in operation.
C	WARNING	The indicator light turns on when the DC breaker is not turned on.
D	FAULT	The indicator light turns on when the system fault has occurred.
E	EMERGENCY	Press the button to disconnect the high-voltage circuit of the system.

6.3 System Powering-Off

STEP 1: Send a “Stop“ command by EMS.

STEP 2: Turn off the AC and DC breakers on the control box.



Figure 6-7: AC and DC Breaker in the OFF Position

STEP 3: Turn the PCS isolation switch to the OFF position (counterclockwise).



Figure 6-8: PCS Isolation Switch in the OFF Position

STEP 4: Turn the liquid cooling power breaker to the OFF position.



Figure 6-9: Liquid Cooling Power Breaker in the OFF Position

STEP 5: Switch the MCCB (QF0) to the OFF position.

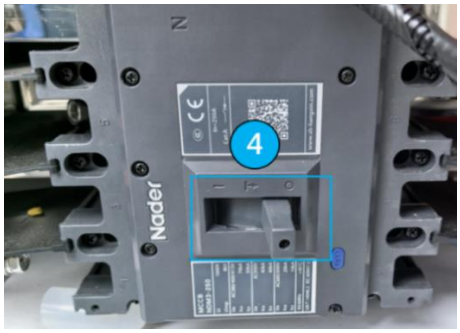


Figure 6-10: MCCB (QF0) in the OFF Position

6.4 Emergency Stop

In the event of an emergency, follow the steps below:

STEP 1: Press the **EMERGENCY** button to disconnect the high-voltage circuit.

The system will stop operating.

STEP 2: Disconnect the external power source.

7. MAINTENANCE GUIDE

7.1 Before Maintenance

⚠ DANGER

- Before performing maintenance operations such as system cleaning, electrical connection inspection, or grounding reliability checks, the system must be powered off. Ensure that no voltage remains in the system before proceeding with any operation.
- The MSD shall not be switched under load. Always isolate and de-energize the circuit before operating the MSD.

⚠WARNING

- Maintenance shall only be carried out by trained and qualified personnel. During maintenance, safety regulations must be strictly followed, and appropriate tools and personal protective equipment (PPE) shall be used.
- It is strictly prohibited to perform maintenance during adverse weather conditions such as thunderstorms, high humidity, or strong winds to ensure the safety of personnel and equipment.
- After any planned or unplanned maintenance or repair, perform insulation resistance tests on all relevant circuits. The system shall be re-energized only after confirming that the insulation condition meets the required standards.

⚠CAUTION

- **DO NOT** use solvents such as water, alcohol, or oil to clean electrical components inside or outside of the equipment.

7.2 Recommended Tools

Table 7-1: Recommended Tool List

Item	Tool	Description
1	Socket and torque wrench	For loosening or tightening bolts
2	Screwdriver	For disassembling components
3	Marker pen	For torque marking
4	Fire detector tester	For testing fire detectors
5	Insulation resistance tester	For measuring insulation resistance
6	Multimeter	For checking voltage and grounding resistance
7	Coolant tester	For testing coolant quality
8	Vacuum cleaner	For removing dust
9	Personal protective equipment (PPE)	Including but not limited to safety helmet, insulated gloves, and insulated shoes

7.4 Key Component List



The design life is provided for maintenance planning reference only. It does not guarantee that no failure will occur during this period, nor does it indicate that replacement is mandatory upon reaching the specified duration. The actual replacement time shall be determined based on the equipment operating condition and inspection results.

Table 7-2: Key Component List

Item	Component	QTY	Design Life
1	Dehumidifier	1	5 years
2	Power Meter	1	10 years
3	Explosion Relief Valve	1	10 years
4	Energy Meter CT	1	10 years
5	EPO Button	1	Depending on actual operating conditions (Electrical life: 100,000 operations)
6	Flood Sensor	1	5 years
7	T/H Detector	1	5 years
8	Door Sensor	1	Depending on actual operating conditions (Electrical life: 5,000,000 operations)
9	Indicator Light	4	5 years
10	Aerosol Fire Extinguisher	1	15 years
11	Smoke Detector	1	10 years
12	Temperature Detector	1	10 years
13	Horn & Strobe	1	10 years
14	Gas Detector	1	3 years
15	Gas-venting Valve (Outlet)	1	15 years
16	Gas-venting Valve (Inlet)	1	15 years
17	Liquid Cooling Unit	1	10 years
18	Control Box	1	15 years
19	Battery Pack	5	Depending on actual operating conditions
20	Switch	1	15 years
21	PCS Module	1	Depending on actual operating conditions
22	MCCB	1	Depending on actual operating conditions (Electrical life: 4,000 operations)
23	Controller	1	5 years
24	Coolant	/	2 years

7.5 SOC Calibration

It is recommended to perform SOC calibration at least once a month; if conditions permit, performing it every two weeks is preferable. Two calibration methods are provided below:

NOTICE

Under normal system operating conditions, if the system completes one full charge within a month, it can be regarded as having completed one SOC calibration.

Method 1 — Full Charge Calibration

STEP 1: All cell voltages are below 3.45V.

STEP 2: Charge the battery to full according to the BMS charging logic and stop. When the highest cell voltage reaches the full-charge threshold and the average voltage exceeds 3.38V, full-charge calibration will be triggered.

Method 2 — OCV Calibration

STEP 1: Power on at low voltage, and ensure the system is in idle state.

STEP 2: All cell voltages shall be within the range of 2.5–3.1V, the minimum battery temperature shall be greater than 10 °C, and the absolute current value shall be less than 5 A.

STEP 3: Keep the system idle for more than 1.5 hours to trigger OCV calibration.

7.6 Maintenance Plan

NOTICE

The following is the recommended maintenance plan. The actual maintenance interval should be adjusted according to the installation environment, as factors such as installation location and site conditions may affect the maintenance cycle. If the operating environment is dusty or prone to sand accumulation, the maintenance interval should be shortened and the maintenance frequency increased accordingly.

Table 7-3: Maintenance Checklist

Frequency	Category	Item
6 months	Fire Protection System	1-1. Ensure the detectors have no visible damage or abnormal conditions.
		1-2. Verify that the detector inspection LED is flashing properly.
		1-3. Use smoke, heat and gas detector testers to verify detector response. **⚠WARNING** Before testing, disconnect the aerosol spraying signal cable to avoid accidental discharge.
		1-4. Check if the aerosol fire extinguisher is within the validity period.
	Coolant	2-1. Take a small amount of coolant from the filling port to check for dirt, sediment, or algae; use a coolant tester to measure pH value and electrolyte concentration. ① pH value shall be within 7.5–9.5 ② Electrolyte concentration: 50% ethylene glycol.
	Cabinet	3-1. Ensure cabinet ventilation openings are free from dust accumulation.
1 year	Cabinet	3-2. Ensure there is no paint peeling, corrosion, or mold.
		3-3. Verify cabinet door lock is intact.
		3-4. Check that sealing strips are securely fixed.
		3-5. Ensure warning labels and nameplates are clearly visible and undamaged.
		3-6. Verify gas venting valves and explosion relief valve are not covered by any objects.
		3-7. Ensure there are no potential hazards or signs of contamination in the surrounding area

1 year	Cabinet	3-8. Verify cabinet cable entry/exit holes are properly sealed.
		3-9. Test the emergency stop button.
		3-10. Confirm cabinet indicator lights are functioning properly.
		3-11. Ensure power cable connections are securely fastened.
	Junction Box	4-1. Verify junction box door lock is intact.
		4-2. Ensure there is no paint peeling, corrosion or mold.
		4-3. Check that sealing strips are securely fixed.
		4-4. Ensure junction box mounting bolts are securely fastened.
		4-5. Check that junction box cable entry/exit holes are properly sealed.
		4-6. Ensure power cable connections are securely fastened.
	Metal Base	5-1. Verify metal base mounting bolts are not loose.
		5-2. Ensure there is no paint peeling, corrosion, or mold.
	Grounding	6-1. Verify system grounding is reliable, and measure the grounding resistance to ensure it doesn't exceed 4Ω.
	Battery Pack	7-1. Check that all cable connections are not loose or disconnected.
		7-2. Ensure there is no visible damage to the cables.
		7-3. Ensure there is no coolant leakage at the cooling plates or at the inlet/outlet ports of the battery packs.
		7-4. Ensure no battery pack emits any burnt or abnormal odor.
		7-5. During system operation, check on the EMS whether all cell temperatures and voltages are displayed normally.

1 year	PCS Module	8-1. Check that all cable connections are not loose or disconnected.
		8-2. Ensure there is no visible damage to the cables.
		8-3. Ensure there is no coolant leakage at the PCS inlet/outlet ports.
		8-4. Verify that the indicator lights on the PCS are functioning normally.
		8-5. During system operation, check on the EMS whether PCS communication is functioning properly.
		8-6. Perform insulation resistance (R/S/T to PE); the insulation resistance shall not be less than 1.8 MΩ.
	Liquid Cooling Pipeline	9-1. Ensure there is no leakage in the liquid cooling pipelines.
		9-2. Ensure there is no blockage in the liquid cooling pipelines.
	Liquid Cooling Unit	10-1. Ensure that the air inlets and outlets of the liquid cooling unit are free from dust accumulation.
		10-2. Verify that the liquid cooling unit is operating without abnormal vibration or noise.
		10-3. Check whether the inlet and outlet water pressures of the liquid cooling unit are within the specified range.: ① Static water pressure — supply pressure: 1.2-3 bar; return pressure: 1.2-3 bar ② Dynamic water pressure — supply pressure: 1.7-3.5 bar; return pressure: 1.2–2.7 bar

7.7 Ventilation Louver Cleaning

⚠ DANGER Power off the system and verify zero voltage before proceeding.

7.7.1 Tool Preparation

Item	Tool	Sourcing
1	Socket and torque wrench	Prepared by user
2	Water Gun	Prepared by user

7.7.2 Inlet Ventilation

STEP 1: Use the key attached to the cabinet door to release the fastener of the front louver. With both hands holding the sides of the louver, remove the entire assembly.

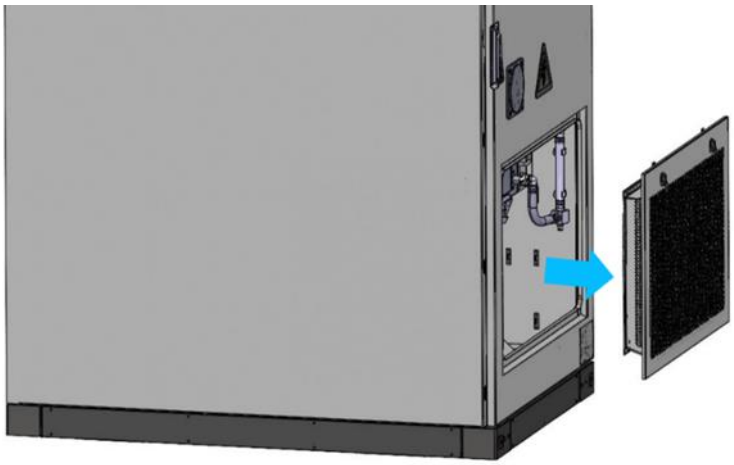


Figure 6-1: Louver Removal

STEP 2: Remove the M5 screws on the filter mesh, and sperate the filter mesh, filter pad and louver.

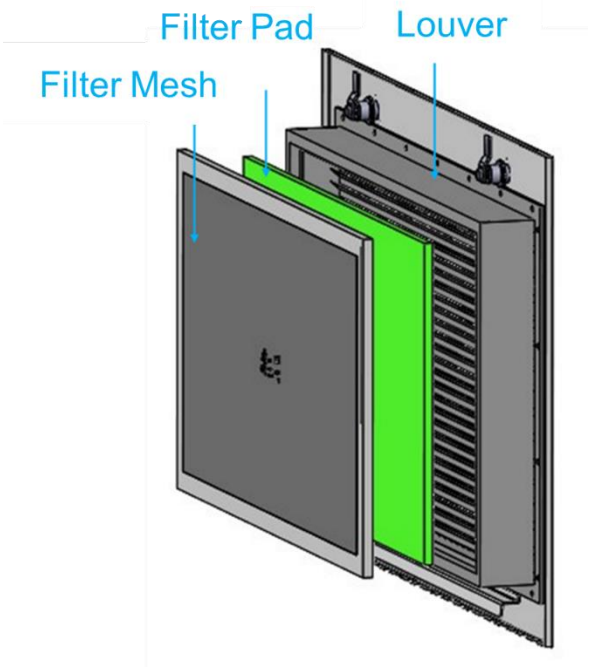


Figure 6-2: Assembly Separation

STEP 3: Use a water spray gun to clean the filter mesh, filter pad, and louver grille.

STEP 4: After the components are fully dried, reinstall them in their original positions:

① Stack in the order: filter mesh → filter pad → louver, and secure the filter mesh using the M5 screws (Torque: 5±0.5 N·m).

② Align the assembled louver with the cabinet window slot and place it in position. After confirming that the louver is fully seated against the cabinet, use the key to tighten it.

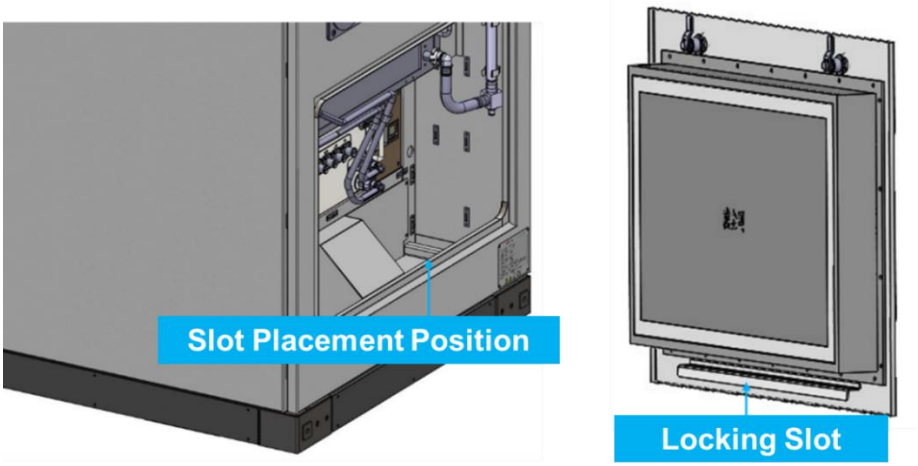


Figure 6-3：Louver Installation

7.7.3 Outlet Ventilation

STEP 1: Remove all screws on the rear louver and take off the entire louver.

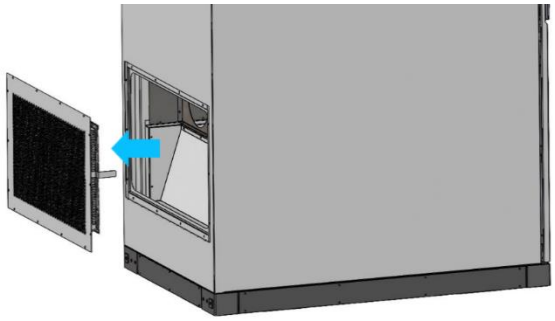


Figure 6-4: Louver Removal

STEP 2: Remove the M5 screws son the filter mesh and separate the filter mesh, filter pad from the louver.

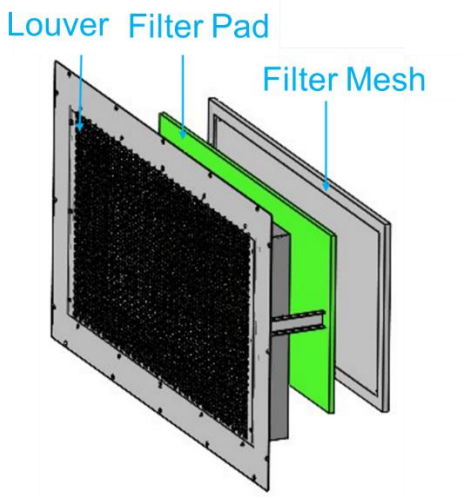


Figure 6-5: Assembly Separation

STEP 3: Use a water spray gun to clean the filter mesh, filter pad and the louver grille.

STEP 4: After all components are fully dried, reinstall them in their original positions:

① Install the filter mesh onto the back side of the louver and secure it with M5 screws (Torque: 5 ± 0.5 N·m).

② Reinstall the assembled louver onto the cabinet and fasten it with the M5 screws in a diagonal sequence (Torque: 5 ± 0.5 N·m).

7.8 Coolant Replacement

⚠ DANGER

Before performing any maintenance work, including system cleaning, coolant replacement, inspection of electrical connections, or grounding reliability checks, the system must be powered down and confirmed to be voltage-free.

7.8.1 Tool Preparation

Item	Tool	Sourcing
1	Liquide cooling display panel and comm. cables	Procured from Delta
2	Filling Pump: Diaphragm self-priming pump with a head of 15–20 m	Prepared by user
3	Fluid Transfer Hoses: ① PCS-side exhaust port: hose with 10 mm inner diameter ② Battery-side exhaust port: hose with 8 mm outer diameter ③ Refill port with SAE18 quick connector: EPDM hose with 17.8 mm inner diameter and 4 mm wall thickness	Prepared by user
4	Coolant (30L / 50% ethylene glycol–water solution / freezing point ≤ −35 °C / suitable for energy storage products) **⚠ CAUTION** **DO NOT** mix coolants, as this may cause crystallization and affect system performance. If a different brand is required, thoroughly flush the piping with deionized water and dry it completely before refilling.	Procured from Delta
5	Coolant container (>30 L)	Prepared by user

7.8.2 Draining

PCS Side

STEP 1: Connect one exhaust hose and one drain hose to the exhaust port and refill port on the PCS side, respectively. Place the other ends of both hoses at the bottom of the coolant container.



Figure 6-6: PCS-side Exhaust Port and Refill Port

STEP 2: Open the blue valves on the PCS-side exhaust port and refill port *(rotate the valves 90°, the horizontal position indicates the open state)*; allow the liquid to drain out naturally.

STEP 3: Disconnect the PCS inlet hose to completely drain the coolant.

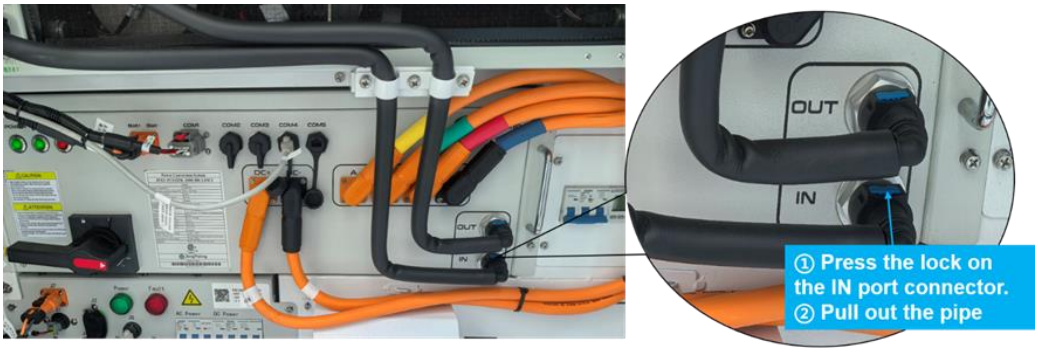


Figure 6-7: PCS IN Port

STEP 4: After draining is completed, reconnect the PCS inlet hose to the IN port. Close the exhaust and refill port valves on the PCS side *(rotate the valves 90°, the vertical position indicates the closed state)*.

Battery Side

STEP 1: Connect one exhaust hose and one drain hose to the exhaust port and refill port on the battery side, respectively. Place the other ends of both hoses at the bottom of the coolant container.

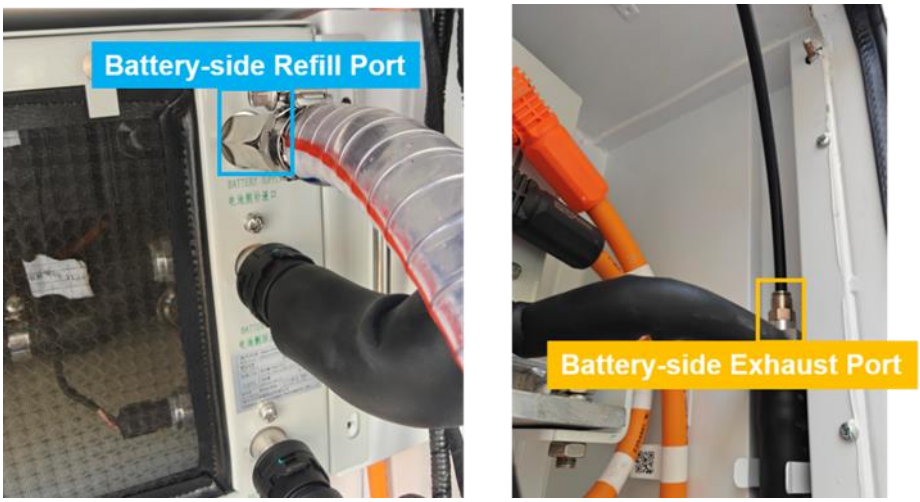


Figure 6-8: Battery-side Exhaust Port and Refill Port

STEP 2: Connect another drain hose to the coolant drain port, which is located at the lower right corner inside the cabinet. Place the other end of the hose at the bottom of the coolant container.

STEP 3: Open the blue valve on the battery-side refill port *(rotate the valve 90°, the horizontal position indicates the open state)*; allow the liquid to drain out naturally.

STEP 4: After draining is completed, close the valve on the battery-side refill port *(rotate the valve 90°, the vertical position indicates the closed state)*.



Figure 6-9: Coolant Drain Port



Figure 6-10: Battery-side Refill Valve – Closed

7.8.3 Filling

NOTICE Close the MCCB (QF0) and the liquid cooling unit power switch, and keep the AC and DC power supply switches in the OFF position.

STEP 1: Connect the communication cable of the liquid-cooling display panel to the debug port of the liquid cooling unit.



Figure 6-11: Debug Port of Liquid Cooling Unit

STEP 2: Power on the MCCB (QF0) and the liquid cooling unit.



Figure 6-12: Liquid Cooling Power Breaker in the ON Position

STEP 3: Connect the piping.



Figure 6-13: PCS and Battery Exhaust Ports and Refill Ports

PCS Side

① Connect the outlet hose of the filling pump to the refill port on the PCS side. Place the inlet hose of the filling pump into the coolant container.

② Connect an exhaust hose to the exhaust port on the PCS side. Place the other end of the hose into the coolant container.

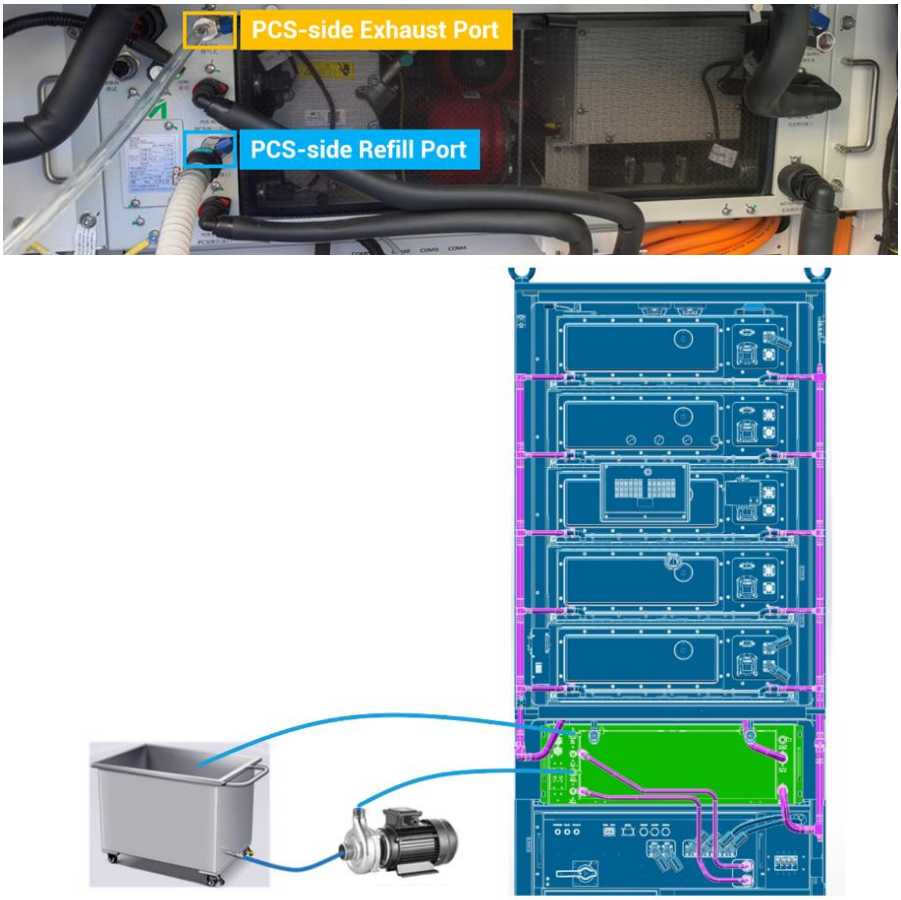


Figure 6-14: PCS-side Hose Connection Illustration

Battery Side

① Connect the outlet hose of the filling pump to the refill port on the battery side. Place the inlet hose of the filling pump into the coolant container.

② Connect an exhaust hose to the exhaust port on the battery side. Place the other end of the hose into the coolant container.

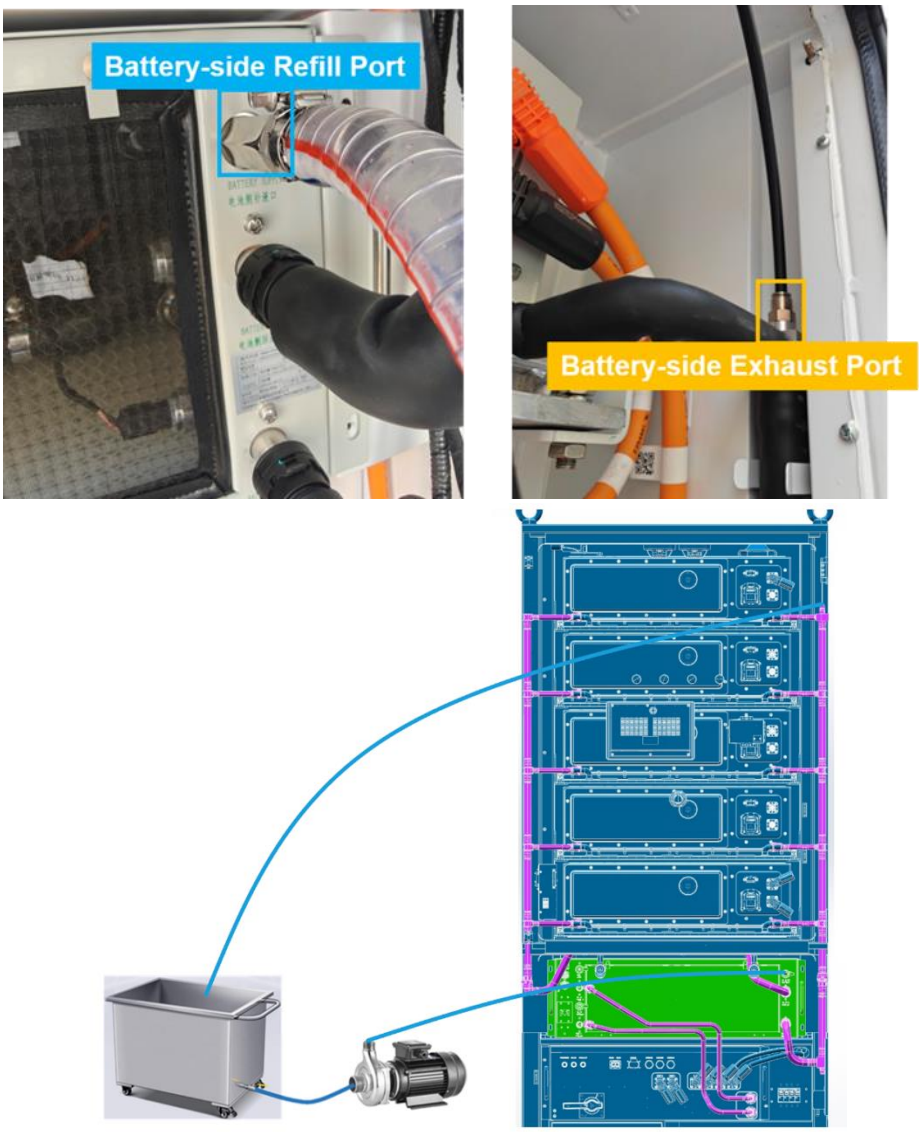


Figure 6-15: Battery-side Hose Connection Illustration

STEP 4: Open the valves *(rotate the valves 90°, the horizontal position indicates the open state)*.

PCS Side

Open the exhaust valve and the refill valve on the PCS side.

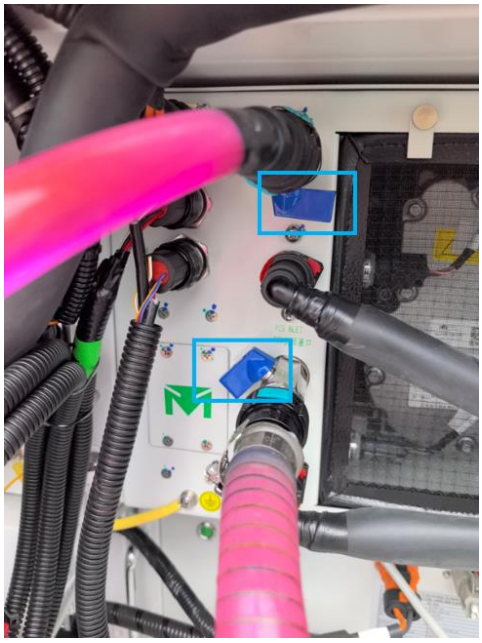


Figure 6-16: PCS-side Valves - Closed

Battery Side

Open the refill valve on the battery side.

(The exhaust pipe will open the valve automatically when inserted into the exhaust port)



Figure 6-17: Battery-side Valve - Open

STEP 5:

PCS Side

① On the liquid-cooling display, navigate to the PCS exhaust mode.
② Turn on the power supply of the filling pump.
③ Observe the exhaust hose: when no air bubbles are visible in the hose, the air has been fully discharged. Then, close the exhaust port valve.

Battery Side

① On the liquid-cooling display, navigate to the battery exhaust mode.
② Turn on the power supply of the filling pump.
③ Observe the exhaust hose: when no air bubbles are visible in the hose, the air has been fully discharged. Then, remove the exhaust pipe to close the exhaust valve.

STEP 6: Observe the pressure value on the liquid-cooling display panel: when the inlet and outlet pressures reach 2.0-2.5 bar, stop refilling.

① Turn off the power supply of the filling pump.
② Close the refill valve *(rotate the valve 90°, the vertical position indicates the closed state)*.

STEP 7: Restore the system.

① Disconnect and remove the filling pump.
② Reinstall the dust caps on the refill and exhaust ports.

8. Emergency Response Measures

In case of any hazardous incident on site, first ensure personnel safety, immediately contact the responsible person or relevant department, and subsequently inform Delta Electronics.

8.1 Coolant Leakage



The coolant is toxic and corrosive to a certain extent. Personnel handling it must wear safety goggles and chemical-resistant gloves.

Minor Leakage

STEP 1: Immediately power down the system.

STEP 2: Use absorbent pads or dry cloths to wipe off the spilled liquid. Place all coolant-contaminated materials in a waste container and handle them according to local regulations.

STEP 3: Carefully inspect all pipeline joints, valves, and surfaces of the liquid-cooling plates for any signs of leakage.

① If a leakage point is found, temporarily seal the leaking area and contact Delta Electronics or Delta-authorized service personnel for on-site handling.

② If no leakage trace is found, and only after confirming it is safe to proceed, briefly run the liquid cooling unit in manual mode via the control panel to help locate the leakage source at hoses, joints, or cooling plates. Observe the equipment carefully and stop the operation immediately if any abnormal condition occurs.

③ If the leakage source still cannot be identified, contact Delta or Delta-authorized service personnel for assistance.

Major Leakage

STEP 1: Immediately power down the system.

STEP 2: Place a container under the leakage point to prevent the coolant from spreading to other areas inside the cabinet. If necessary, drain the system coolant to avoid further risk.

STEP 3: Contact Delta Electronics or Delta-authorized service personnel for on-site handling.

STEP 4: Use absorbent pads or dry cloths to wipe off the spilled liquid. Place all coolant-contaminated materials in a waste container and handle them according to local regulations.

8.2 Fire Response

⚠ DANGER

- **DO NOT** open the cabinet doors under any circumstances without authorization.
- When the battery temperature becomes excessively high, it may cause deformation, damage, or electrolyte leakage, releasing toxic gases. In such cases, personnel must wear respiratory protective equipment and must not approach the burning area to avoid skin irritation or chemical burns.

NOTICE

- Use CO_2 or dry powder extinguishers to extinguish the fire.

In the event of a fire, follow the procedures below:

STEP 1: Evacuate the equipment area immediately and activate the fire alarm. At the same time, call the fire department and provide relevant product information to the emergency responders.

STEP 2: If conditions permit and personal safety can be ensured, remotely shut down the system power supply.

STEP 3: After the fire has been completely extinguished and confirmed safe by professional firefighters, follow local regulations and allow qualified personnel to handle post-fire treatment.

STEP 4: Contact Delta Electronics for inspection and evaluation.